

var, x, y, z, w
label, L, Next, Done
lit
f
dsort, D
tvar, a, b
index, i, j, k, n

<i>terminals</i>	$::=$ $\rightarrow$ $\rightsquigarrow$ $\rightarrow$ $ $ $\vee$ $@$ $\rightarrow \circ$ $\wedge$ $\Rightarrow$ $\in$ <code>true</code> <code>false</code> $\top$ $\perp$ $\forall$ <code>is</code> <code>let</code> <code>res</code> <code>core</code> <code>log</code> <code>auto</code> <code>return</code> <code>take</code> <code>fail</code> <code>case</code> <code>iftrue</code> <code>iffalse</code> <code>unfold</code> <code>annot</code> <code>do</code> <code>.</code>	
<i>blit</i>	$::=$ $ $ <code>true</code> $ $ <code>false</code>	boolean literals
<i>eff</i> , ϵ	$::=$ $ $ 0 $ $ 1 $ $ S	effects
<i>kind</i>	$::=$ $ $ <code>type</code> $ $ <code>sort</code>	kinds
τ , <i>A</i> , <i>B</i> , <i>C</i>	$::=$ $ $ <code>Int</code> $ $ <code>Bool</code> $ $ <code>Ptr</code> τ' $ $ $(\tau_1, \dots, \tau_n)$ $ $ $D(\tau_1, \dots, \tau_n)$	sorts integers booleans pointer tuple datatype application

	$\begin{array}{l} \text{Pred } \tau' \\ a \\ [\gamma]\tau \end{array}$	predicate sort type variable	M
$\overline{A}$	$\begin{array}{l} ::= \\ \square \\ A :: \overline{A} \end{array}$	type lists	
F	$\begin{array}{l} ::= \\ \tau \rightarrow_{\text{eff}} \tau' \end{array}$	function signatures	
Γ	$\begin{array}{l} ::= \\ \cdot \\ \Gamma, x :_{\text{eff}} \tau \\ \Gamma, a : \textit{kind} \\ \Gamma, \Gamma' \\ \Delta \end{array}$	contexts	M
γ	$\begin{array}{l} ::= \\ \cdot \\ \gamma, ce_1/x_1, \dots, ce_n/x_n \\ \gamma, \tau_1/a_1, \dots, \tau_n/a_n \\ \text{id}(\Gamma) \\ \gamma; \gamma' \end{array}$	explicit substitutions	M M
Σ_c	$\begin{array}{l} ::= \\ \cdot \\ \Sigma_c, f : F \\ \Sigma_c, \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = ce \\ \Sigma_c, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} \\ \Sigma_c, \text{type } D(\overline{a}) = \{\overline{L : A}\} \end{array}$	core signatures	
Σ_s	$\begin{array}{l} ::= \\ \cdot \\ \Sigma_s, f : F \\ \Sigma_s, \text{fun } f : \tau \rightarrow_{\text{eff}} \tau' = \{\overline{\text{pat} \rightarrow \text{se}}\} \\ \Sigma_s, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} \\ \Sigma_s, \text{type } D(\overline{a}) = \{\overline{L : A}\} \end{array}$	surface signatures	
Σ_r	$\begin{array}{l} ::= \\ \cdot \\ \Sigma_r, f : F \\ \Sigma_r, \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = ce \\ \Sigma_r, f : RF \\ \Sigma_r, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} \\ \Sigma_r, \text{type } D(\overline{a}) = \{\overline{L : A}\} \end{array}$	refined signatures	
$\textit{prim}$	$\begin{array}{l} ::= \\ \text{Add} \\ \text{Sub} \\ \text{Mul} \\ \text{Div} \\ \text{And} \end{array}$	primitives addition subtraction multiplication division conjunction	

	$ \begin{array}{l} \text{fun } f : \tau \rightarrow_{\text{eff}} \tau' = \{\overline{\text{pat}} \rightarrow \text{se}\} \text{ prog} \\ \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ prog} \\ \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{ prog} \\ \text{main} : \tau_{\text{eff}} = \text{se} \end{array} $	
<i>cprog</i>	$ \begin{array}{l} ::= \\ \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = \text{ce} \quad \text{cprog} \\ \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ cprog} \\ \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{ cprog} \\ \text{main} : \tau_{\text{eff}} = \text{ce} \end{array} $	core programs
<i>rprog</i>	$ \begin{array}{l} ::= \\ \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ rprog} \\ \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{ rprog} \\ \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = \text{ce} \quad \text{rprog} \\ \text{fun } f(q) : Pf_1 \multimap Pf_2 [\text{eff}] = \text{crt} \quad \text{rprog} \\ \text{main} : Pf_{\text{eff}} = \text{crt} \end{array} $	refined programs
<i>pat</i>	$ \begin{array}{l} ::= \\ x \\ L \text{ pat} \\ (\text{pat}_1, \dots, \text{pat}_n) \end{array} $	patterns
β	$ \begin{array}{l} ::= \\ \cdot \\ \text{pat} : \tau, \beta \\ \text{pat}_1 : \tau_1, \dots, \text{pat}_n : \tau_n, \beta \\ \text{pat} : \tau \end{array} $	pattern bindings
$\mathcal{E}$	$ \begin{array}{l} ::= \\ [\cdot] \\ \text{let } x = \text{ce}; \mathcal{E} \end{array} $	evaluation contexts
$\overline{y}$	$ \begin{array}{l} ::= \\ \cdot \\ y, \overline{y} \\ y_1, \dots, y_n, \overline{y} \end{array} $	scrutinee variables
<i>M</i>	$ \begin{array}{l} ::= \\ \cdot \\ \beta \rightarrow [\Gamma][\mathcal{E}] \text{se} \mid M \\ \beta \rightarrow [\Gamma][\mathcal{E}] \text{se} \end{array} $	match branches
<i>formula</i>	$ \begin{array}{l} ::= \\ \text{judgement} \\ x :_{\text{eff}} \tau \in \Gamma \\ f : \tau \rightarrow_{\text{eff}} \tau' \in \Sigma_c \\ A \text{ eqtype} \\ \text{sort } D(a_1, \dots, a_i) \text{ in } \Sigma_c \\ L : \tau \in \text{ctors}(D) \in \Sigma_c \\ \text{sort } D(a_1, \dots, a_k) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ in } \Sigma_c \\ \text{type } D(a_1, \dots, a_i) \text{ in } \Sigma_c \\ L : A \in \text{tyctors}(D) \in \Sigma_c \end{array} $	

$\mathbf{type} \ D(a_1, \dots, a_k) = \{L_1 : A_1 \mid \dots \mid L_n : A_n\} \mathbf{in} \ \Sigma_c$
 $\mathbf{type} \ D(a, b) = \{\mathbf{Next}:a \mid \mathbf{Done}:b\} \in \Sigma_c$
 $a:kind \in \Gamma$
 $kind \leq kind'$
 $L \neq L'$
 $M = \overline{pat:\tau, \cdot \rightarrow [\cdot][\cdot] \ se}$
 $ce = \mathcal{E}[ce']$
 $D \neq D'$
 $x \neq y$
 $a \neq b$
 $n = k$
 $eff = eff_1 \vee eff_2$
 $eff = eff_1 \vee eff_2 \vee eff_3$
 $eff = eff_1 \vee \dots \vee eff_n$
 $\Gamma = x_1 : \tau_1[eff_1], \dots, x_n : \tau_n[eff_n]$
 $\Gamma = a : \bar{k}$
 $formula_1 \dots formula_n$
 $x:ce \ [\mathbf{log}] \in \Delta$
 $x:ce@ce' \ [\mathbf{res}(u)] \in \Delta$
 $x:eff\tau \in \Delta$
 $u \in \{1, ?\}$
 $u \in \{0, ?\}$
 $u = u_1 \sqcap u_2$
 $\Delta' = \mathbf{affinize}(\Delta)$
 $\mathbf{zero}(\Delta)$
 $\Gamma = |\Delta|$
 $\Gamma = \mathbf{Bind}(\cdot; Pf)$
 $\mathcal{C} = \Delta \Rightarrow \mathcal{C}'$
 $f(\mathbf{x} : \tau) \rightarrow_{eff} \tau' = ce \in \Sigma_c$
 $eff' \in \{\mathbf{S}, \lfloor eff \rfloor\}$
 $f:RF \in \Sigma_c$
 $f:RF \in \Sigma_r$
 $f:\tau \rightarrow_{eff} \tau' \in \Sigma_r$
 $f(\mathbf{x} : \tau) \rightarrow_{eff} \tau' = ce \in \Sigma_r$
 $\mathbf{len}(\Delta) = \mathbf{len}(\Delta')$
 $\Delta = \Delta'$
 $A = \mathbf{Prod}(\overline{A})$
 $prim:RF$
 $ce = \mathbf{branch}(L, x, ce')$

u ::=
 $\mid$ 0
 $\mid$ 1
 $\mid$?

usage flags

Δ ::=
 $\mid$.
 $\mid$ $x_1 : \tau_1[eff_1], \dots, x_n : \tau_n[eff_n]$
 $\mid$ $x_1 : ce_1[\mathbf{log}], \dots, x_n : ce_n[\mathbf{log}]$
 $\mid$ $x_1 : ce_1@ce'_1[\mathbf{res}(u_1)], \dots, x_n : ce_n@ce'_n[\mathbf{res}(u_n)]$
 $\mid$ Δ, Δ'

refined contexts

$\overline{\Delta}$::=

context lists

	$ \begin{array}{ l} \Delta; \overline{\Delta} \\ \cdot \\ (\overline{\Delta}) \\ \Delta_1; \dots; \Delta_n \end{array} $	
$\mathcal{C}$	$ \begin{array}{ l} ::= \\ \top \\ \mathcal{C}_1 \wedge \mathcal{C}_2 \\ \forall x_1 : \tau_1 \dots \forall x_n : \tau_n. \mathcal{C} \\ ce \Rightarrow \mathcal{C} \\ ce \\ \perp \\ (\mathcal{C}) \\ \text{is } ce \ L \end{array} $	SMT constraints
Pf	$ \begin{array}{ l} ::= \\ \cdot \\ x :_{eff} \tau, Pf \\ ce \ [\text{log}], Pf \\ ce @ ce' \ [\text{res}], Pf \\ (y). ce \ [\text{res}], Pf \\ [\gamma] Pf \end{array} $	proof sorts
RF	$ \begin{array}{ l} ::= \\ Pf_1 \multimap Pf_2 \ [eff] \end{array} $	refined function types
$cpat$	$ \begin{array}{ l} ::= \\ x \\ (cpat_1, \dots, cpat_n) \end{array} $	core patterns
$lpat$	$ \begin{array}{ l} ::= \\ x \\ \text{auto} \end{array} $	logical patterns
$rpat$	$ \begin{array}{ l} ::= \\ x \\ \text{return } lpat \\ \text{take } (cpat, rpat_1); rpat_2 \\ \text{fail } [lpat] \\ \text{let } [lpat] cpat; rpat \\ \text{case } [lpat] L \ cpat; rpat \\ \text{if true}; rpat \\ \text{if false}; rpat \\ \text{unfold}; rpat \\ \text{annot}; rpat \end{array} $	resource patterns
rpb	$ \begin{array}{ l} ::= \\ \text{do } cpat = rpat \\ rpat \end{array} $	resource pattern bases
q	$ \begin{array}{ l} ::= \\ \cdot \\ (cpat, q) \end{array} $	refined patterns

	$\begin{array}{l} \quad (\log \textit{lpat}, q) \\ \quad (\textit{res rpb}, q) \end{array}$	
$\textit{crt}$	$\begin{array}{l} ::= \\ \quad \text{let } q = \textit{crt}_1; \textit{crt}_2 \\ \quad \text{let } \log \textit{lpat} = \textit{lpf}; \textit{crt} \\ \quad \text{let } \textit{res rpat} = \textit{rpf}; \textit{crt} \\ \quad \text{let core } [\textit{lpat}] \textit{cpat} = \textit{ce}; \textit{crt} \\ \quad \textit{crt} : \textit{Pf} \\ \quad \textit{prim rsp} \\ \quad \textit{f rsp} \\ \quad (\textit{rsp}) \\ \quad \textbf{iter } [\textit{ce}] (q = \textit{crt}_1) \{ \textit{crt}_2 \} \\ \quad \textbf{if } [x] \textit{ce} \textbf{ then } \textit{crt}_1 \textbf{ else } \textit{crt}_2 \\ \quad \textbf{case } [y] \textit{ce} \textbf{ of } \{ L_1 x_1 \rightarrow \textit{crt}_1 \mid \dots \mid L_n x_n \rightarrow \textit{crt}_n \} \\ \quad \textbf{exfalse} \\ \quad \textbf{open-take } \textit{rpf} \end{array}$	core refined terms
$\textit{rsp}$	$\begin{array}{l} ::= \\ \quad \cdot \\ \quad \textit{ce}, \textit{rsp} \\ \quad \textit{rpf}, \textit{rsp} \\ \quad \textit{lpf}, \textit{rsp} \end{array}$	refined spines
$\textit{lpf}$	$\begin{array}{l} ::= \\ \quad x \\ \quad \textbf{auto} \\ \quad \textbf{unfold } f(\textit{ce}) \\ \quad \textbf{open-ret } \textit{rpf} \\ \quad \textit{lpf} : \textit{ce} \end{array}$	logical proof facts
$\textit{rpf}$	$\begin{array}{l} ::= \\ \quad x \\ \quad \textit{rpf} : \textit{ce} @ \textit{ce}' \\ \quad \textbf{return } \textit{lpf} \\ \quad \textbf{take } (\textit{rpf}_1, \textit{rpf}_2) \\ \quad \textbf{fail } [\textit{lpf}] \\ \quad \textbf{let } [\textit{lpat}] \textit{cpat}; \textit{rpf} \\ \quad \textbf{case } [\textit{lpat}] L \textit{cpat}; \textit{rpf} \\ \quad \textbf{iftrue}; \textit{rpf} \\ \quad \textbf{iffalse}; \textit{rpf} \\ \quad \textbf{unfold}; \textit{rpf} \\ \quad \textbf{annot}; \textit{rpf} \end{array}$	resource proof facts
$\textit{Jtype}$	$\begin{array}{l} ::= \\ \quad \textit{kind} \leq \textit{kind}' \\ \quad \Sigma_c; \Gamma \vdash \tau : \textit{kind} \\ \quad \Sigma_c; \Gamma; D'(\bar{a}) \vdash, \textbf{guarded} \\ \quad \Sigma_c \vdash \textbf{sort } D(\bar{a}) = \{ \overline{L : \tau} \} \textbf{ ok} \\ \quad \Sigma_c \vdash \textbf{type } D(\bar{a}) = \{ \overline{L : A} \} \textbf{ ok} \\ \quad \Sigma_c; \Gamma \vdash \gamma : \Gamma' \\ \quad \gamma = \textbf{id}(\Gamma) \\ \quad \tau' = [\gamma] \tau \end{array}$	

	$ \begin{array}{l} \quad ce' = [\gamma]ce \\ \quad Pf' = [\gamma]Pf \\ \quad \gamma'' = \gamma ; \gamma' \\ \quad A \text{ eqtype} \\ \quad eff \leq eff' \\ \quad eff' = \lfloor eff \rfloor \\ \quad eff = eff_1 \vee eff_2 \\ \quad prim:\tau \rightarrow_{eff} \tau' \\ \quad \Sigma_c; \Gamma \vdash_{eff_0} ce \Rightarrow \tau \\ \quad \Sigma_c; \Gamma \vdash_{eff_0} ce \Leftarrow \tau \\ \quad \Sigma_c \vdash cprog \end{array} $
$Jcore$	$ \begin{array}{l} ::= \\ \quad L : \tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \\ \quad D(a_1, \dots, a_k) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \in \Sigma_c \end{array} $
$Jelaborate$	$ \begin{array}{l} ::= \\ \quad \Sigma_c; \Gamma \vdash_{eff_0} se \Rightarrow \tau \rightsquigarrow ce \\ \quad \Sigma_c; \Gamma \vdash_{eff_0} se \Leftarrow \tau \rightsquigarrow ce \\ \quad \Sigma_s \vdash prog \\ \quad \rightsquigarrow \Sigma_c \vdash cprog \end{array} $
$Jmatch$	$ \begin{array}{l} ::= \\ \quad M \text{ has_con} \\ \quad M \text{ has_tup} \\ \quad M / var \rightsquigarrow_{eff}^y M' \\ \quad M / L:\tau \rightsquigarrow_{eff}^y M' \\ \quad M / \mathbf{tup} \rightsquigarrow_{eff}^y M' \\ \quad \Sigma_c; \Gamma \vdash_{eff_0} \bar{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow ce \\ \quad \mathcal{E}[ce] \rightsquigarrow ce' \end{array} $
$Jrefine$	$ \begin{array}{l} ::= \\ \quad \Sigma_r \vdash \Delta \text{ wf} \\ \quad \Gamma = \Delta \\ \quad \Delta' = \text{affinize}(\Delta) \\ \quad \text{zero}(\Delta) \\ \quad \mathcal{C} = \Delta \Rightarrow \mathcal{C}' \\ \quad \Sigma_r; \Gamma \vdash \mathcal{C} \text{ wf} \\ \quad \Sigma_r; \Gamma \vdash_{eff} Pf \text{ wf} \\ \quad \Sigma_r \vdash RF \text{ wf} \\ \quad \Sigma_r \vdash f:RF \\ \quad \bar{A} = \{Pf\} \\ \quad \Sigma_c = \text{Comp}(\Sigma_r) \\ \quad \vdash \Sigma_r \text{ wf} \\ \quad \Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r; \Delta \vdash x : ce @ ce' \dashv \Delta' \\ \quad \Sigma_r; \Delta \vdash rpf \Rightarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r; \Delta \vdash rpf \Leftarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta \\ \quad \Sigma_r; \Delta \vdash_{eff} cpat : \tau \dashv \Delta' \rightsquigarrow ce \\ \quad \Sigma_r; \Delta \vdash lpat : ce \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \quad \Sigma_r; \Delta \vdash_{eff} rpat : ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \end{array} $

	$\Sigma_r; \Delta \vdash_{\text{eff}} q : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Rightarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r; \Delta \vdash_{\text{eff}} y:\text{ce} \text{ is } Lx \rightarrow \text{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}$ $\Sigma_r \vdash \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}$ $\Sigma_r \vdash \Box \Delta = \Delta$
<i>judgement</i>	$::=$ $Jtype$ $Jcore$ $Jelaborate$ $Jmatch$ $Jrefine$
<i>user_syntax</i>	$::=$ var $label$ lit f $dsort$ $tvar$ $index$ $terminals$ $blit$ eff $kind$ τ $\overline{A}$ F Γ γ Σ_c Σ_s Σ_r $prim$ ce se $prog$ $cprog$ $rprog$ pat β $\mathcal{E}$ $\overline{y}$ M $formula$ u Δ $\overline{\Delta}$ $\mathcal{C}$ Pf

RF
 $cpat$
 $lpat$
 $rpat$
 rpb
 q
 crt
 rsp
 lpf
 rpf

$kind \leq kind'$

$\frac{}{type \leq type} \quad SK_TYPE_TYPE$
 $\frac{}{type \leq sort} \quad SK_TYPE_SORT$
 $\frac{}{sort \leq sort} \quad SK_SORT_SORT$

$\Sigma_c; \Gamma \vdash \tau:kind$

$\frac{}{\Sigma_c; \Gamma \vdash \text{Int}:kind} \quad KWF_INT$
 $\frac{}{\Sigma_c; \Gamma \vdash \text{Bool}:kind} \quad KWF_BOOL$
 $\frac{\Sigma_c; \Gamma \vdash \tau:type}{\Sigma_c; \Gamma \vdash \text{Ptr } \tau:kind} \quad KWF_PTR$
 $\frac{\Sigma_c; \Gamma \vdash \tau_1:kind \quad \dots \quad \Sigma_c; \Gamma \vdash \tau_n:kind}{\Sigma_c; \Gamma \vdash (\tau_1, \dots, \tau_n):kind} \quad KWF_RECORD$
 $\frac{\mathbf{type } D(a_1, \dots, a_k) \text{ in } \Sigma_c \quad n = k \quad \Sigma_c; \Gamma \vdash \tau_1:type \quad \dots \quad \Sigma_c; \Gamma \vdash \tau_n:type}{\Sigma_c; \Gamma \vdash D(\tau_1, \dots, \tau_n):type} \quad KWF_TYPE_APP$
 $\frac{\mathbf{sort } D(a_1, \dots, a_k) \text{ in } \Sigma_c \quad n = k \quad \Sigma_c; \Gamma \vdash \tau_1:sort \quad \dots \quad \Sigma_c; \Gamma \vdash \tau_n:sort}{\Sigma_c; \Gamma \vdash D(\tau_1, \dots, \tau_n):sort} \quad KWF_SORT_APP$
 $\frac{\Sigma_c; \Gamma \vdash \tau:sort}{\Sigma_c; \Gamma \vdash \text{Pred } \tau:sort} \quad KWF_PRED$
 $\frac{a:kind' \in \Gamma \quad kind' \leq kind}{\Sigma_c; \Gamma \vdash a:kind} \quad KWF_TVAR$

$\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}$

$\frac{}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad TWFG_INT$
 $\frac{}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad TWFG_BOOL$
 $\frac{\Sigma_c, \text{ type } D'(\bar{a}) = \{\bar{L} : \bar{A}\}; \Gamma \vdash \tau:type}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad TWFG_PTR$
 $\frac{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded} \quad \dots \quad \Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad TWFG_RECORD$

$$\frac{\text{type } D(b_1, \dots, b_k) \text{ in } \Sigma_c \quad n = k \quad D \neq D' \quad \Sigma_c; \Gamma \vdash \tau_1 : \text{type} \quad \dots \quad \Sigma_c; \Gamma \vdash \tau_n : \text{type}}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad \text{TWFG_APP}$$

$$\frac{a : \text{kind} \in \Gamma}{\Sigma_c; \Gamma; D'(\bar{a}) \vdash, \text{ guarded}} \quad \text{TWFG_TVAR}$$

$$\boxed{\Sigma_c \vdash \text{sort } D(\bar{a}) = \{\overline{L : \tau}\} \text{ ok}}$$

$$\frac{\Gamma = \overline{a : k} \quad \Sigma_c, \text{sort } D(\bar{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_1 : \text{sort} \quad \dots \quad \Sigma_c, \text{sort } D(\bar{a}) = \{\overline{L : \tau}\}; \Gamma \vdash \tau_n : \text{sort}}{\Sigma_c \vdash \text{sort } D(\bar{a}) = \{\overline{L : \tau}\} \text{ ok}} \quad \text{SDWF_OK}$$

$$\boxed{\Sigma_c \vdash \text{type } D(\bar{a}) = \{\overline{L : A}\} \text{ ok}}$$

$$\frac{\Gamma = \overline{a : k} \quad \Sigma_c; \Gamma; D(\bar{a}) \vdash, \text{ guarded} \quad \dots \quad \Sigma_c; \Gamma; D(\bar{a}) \vdash, \text{ guarded}}{\Sigma_c \vdash \text{type } D(\bar{a}) = \{\overline{L : A}\} \text{ ok}} \quad \text{TDWF_OK}$$

$$\boxed{\Sigma_c; \Gamma \vdash \gamma : \Gamma'}$$

$$\overline{\Sigma_c; \Gamma \vdash \dots} \quad \text{SUBWF_EMPTY}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash_{\text{eff}} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash \gamma, ce/x : \Gamma', x :_{\text{eff}} \tau} \quad \text{SUBWF_TERM}$$

$$\frac{\Sigma_c; \Gamma \vdash \gamma : \Gamma' \quad \Sigma_c; \Gamma \vdash \tau : \text{kind}}{\Sigma_c; \Gamma \vdash \gamma, \tau/a : \Gamma', a : \text{kind}} \quad \text{SUBWF_TYPE}$$

$$\boxed{\gamma = \text{id}(\Gamma)}$$

$$\overline{\cdot = \text{id}(\cdot)} \quad \text{IDDEF_EMPTY}$$

$$\frac{\gamma = \text{id}(\Gamma)}{\gamma, x/x = \text{id}(\Gamma, x :_{\text{eff}} \tau)} \quad \text{IDDEF_TERM}$$

$$\frac{\gamma = \text{id}(\Gamma)}{\gamma, a/a = \text{id}(\Gamma, a : \text{kind})} \quad \text{IDDEF_TYPE}$$

$$\boxed{\tau' = [\gamma]\tau}$$

$$\overline{\text{Int} = [\gamma]\text{Int}} \quad \text{SS_INT}$$

$$\overline{\text{Bool} = [\gamma]\text{Bool}} \quad \text{SS_BOOL}$$

$$\frac{\tau' = [\gamma]\tau}{\text{Ptr } \tau' = [\gamma]\text{Ptr } \tau} \quad \text{SS_PTR}$$

$$\frac{\tau'_1 = [\gamma]\tau_1 \quad \dots \quad \tau'_n = [\gamma]\tau_n}{(\tau'_1, \dots, \tau'_n) = [\gamma](\tau_1, \dots, \tau_n)} \quad \text{SS_RECORD}$$

$$\frac{\tau'_1 = [\gamma]\tau_1 \quad \dots \quad \tau'_n = [\gamma]\tau_n}{D(\tau'_1, \dots, \tau'_n) = [\gamma]D(\tau_1, \dots, \tau_n)} \quad \text{SS_APP}$$

$$\frac{\tau' = [\gamma]\tau}{\text{Pred } \tau' = [\gamma]\text{Pred } \tau} \quad \text{SS_PRED}$$

$$\overline{\tau = [\gamma, \tau/a]a} \quad \text{SS_TVAR_HERE}$$

$$\frac{a \neq b \quad \tau' = [\gamma]b}{\tau' = [\gamma, \tau/a]b} \quad \text{SS_TVAR_SKIP_TY}$$

$$\frac{\tau' = [\gamma]b}{\tau' = [\gamma, ce/x]b} \quad \text{SS_TVAR_SKIP_TM}$$

$$\boxed{ce' = [\gamma]ce}$$

$$\frac{}{ce = [\gamma, ce/x]x} \quad \text{TS_VAR_HERE}$$

$$\frac{x \neq y \quad ce' = [\gamma]x}{ce' = [\gamma, ce/y]x} \quad \text{TS_VAR_SKIP_TM}$$

$$\frac{ce' = [\gamma]x}{ce' = [\gamma, \tau/a]x} \quad \text{TS_VAR_SKIP_TY}$$

$$\frac{}{lit = [\gamma]lit} \quad \text{TS_INTLIT}$$

$$\frac{}{blit = [\gamma]blit} \quad \text{TS_BOOLLIT}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma, x/x]ce_2}{\text{let } x = ce'_1; ce'_2 = [\gamma]\text{let } x = ce_1; ce_2} \quad \text{TS_LET}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad \dots \quad ce'_n = [\gamma]ce_n}{(ce'_1, \dots, ce'_n) = [\gamma](ce_1, \dots, ce_n)} \quad \text{TS_TUPLE}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma, x_1/x_1, \dots, x_n/x_n]ce_2}{\text{let } (x_1, \dots, x_n) = ce'_1; ce'_2 = [\gamma]\text{let } (x_1, \dots, x_n) = ce_1; ce_2} \quad \text{TS_LETTUPLE}$$

$$\frac{ce' = [\gamma]ce}{L \ ce' = [\gamma]L \ ce} \quad \text{TS_INJECT}$$

$$\frac{ce' = [\gamma]ce \quad ce'_1 = [\gamma, x_1/x_1]ce_1 \quad \dots \quad ce'_n = [\gamma, x_n/x_n]ce_n}{\text{case } ce' \text{ of } \{L_1 \ x_1 \rightarrow ce'_1 \mid \dots \mid L_n \ x_n \rightarrow ce'_n\} = [\gamma]\text{case } ce \text{ of } \{L_1 \ x_1 \rightarrow ce_1 \mid \dots \mid L_n \ x_n \rightarrow ce_n\}} \quad \text{TS_CASE}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma, x/x]ce_2}{\text{iter } (x = ce'_1)\{ce'_2\} = [\gamma]\text{iter } (x = ce_1)\{ce_2\}} \quad \text{TS_ITER}$$

$$\frac{ce' = [\gamma]ce}{\text{prim } ce' = [\gamma]\text{prim } ce} \quad \text{TS_PRIM}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma]ce_2 \quad ce'_3 = [\gamma]ce_3}{\text{if } ce'_1 \text{ then } ce'_2 \text{ else } ce'_3 = [\gamma]\text{if } ce_1 \text{ then } ce_2 \text{ else } ce_3} \quad \text{TS_IF}$$

$$\frac{ce' = [\gamma]ce \quad \tau' = [\gamma]\tau}{ce' : \tau' = [\gamma]ce : \tau} \quad \text{TS_ANNOT}$$

$$\frac{ce' = [\gamma]ce}{f \ ce' = [\gamma]f \ ce} \quad \text{TS_CALL}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma]ce_2}{ce'_1 \doteq ce'_2 = [\gamma]ce_1 \doteq ce_2} \quad \text{TS_EQ}$$

$$\frac{ce'_1 = [\gamma]ce_1 \quad ce'_2 = [\gamma, x/x]ce_2}{\text{take } x = ce'_1; ce'_2 = [\gamma]\text{take } x = ce_1; ce_2} \quad \text{TS_TAKE}$$

$$\frac{ce' = [\gamma]ce}{\text{return } ce' = [\gamma]\text{return } ce} \quad \text{TS_RETURN}$$

$$\boxed{Pf' = [\gamma]Pf}$$

$$\begin{array}{c}
\frac{}{\cdot = [\gamma] \cdot} \text{PS_NIL} \\
\\
\frac{\tau' = [\gamma] \tau \quad Pf' = [\gamma, x/x] Pf}{x :_{\text{eff}} \tau', Pf' = [\gamma] x :_{\text{eff}} \tau, Pf} \text{PS_COMP} \\
\\
\frac{ce' = [\gamma] ce \quad Pf' = [\gamma] Pf}{ce' [\log], Pf' = [\gamma] ce [\log], Pf} \text{PS_LOG} \\
\\
\frac{ce'_1 = [\gamma] ce_1 \quad ce'_2 = [\gamma] ce_2 \quad Pf' = [\gamma] Pf}{ce'_1 @ ce'_2 [\text{res}], Pf' = [\gamma] ce_1 @ ce_2 [\text{res}], Pf} \text{PS_RES} \\
\\
\frac{ce' = [\gamma, y/y] ce \quad Pf' = [\gamma] Pf}{(y).ce' [\text{res}], Pf' = [\gamma] (y).ce [\text{res}], Pf} \text{PS_DEPRES}
\end{array}$$

$$\gamma'' = \gamma ; \gamma'$$

$$\begin{array}{c}
\frac{}{\cdot = \gamma ; \cdot} \text{COMPDEF_EMPTY} \\
\\
\frac{\gamma'' = \gamma ; \gamma' \quad ce' = [\gamma] ce}{\gamma'', ce' / x = \gamma ; \gamma', ce / x} \text{COMPDEF_TERM} \\
\\
\frac{\gamma'' = \gamma ; \gamma' \quad \tau' = [\gamma] \tau}{\gamma'', \tau' / a = \gamma ; \gamma', \tau / a} \text{COMPDEF_TYPE}
\end{array}$$

$$A \text{ eqtype}$$

$$\begin{array}{c}
\frac{}{\text{Int eqtype}} \text{EQTY_INT} \\
\\
\frac{}{\text{Bool eqtype}} \text{EQTY_BOOL} \\
\\
\frac{}{\text{Ptr } A \text{ eqtype}} \text{EQTY_PTR}
\end{array}$$

$$\text{eff} \leq \text{eff}'$$

$$\begin{array}{c}
\frac{}{\mathbf{0} \leq \mathbf{0}} \text{SUB_PURE_PURE} \\
\\
\frac{}{\mathbf{0} \leq \mathbf{1}} \text{SUB_PURE_IMPURE} \\
\\
\frac{}{\mathbf{1} \leq \mathbf{1}} \text{SUB_IMPURE_IMPURE} \\
\\
\frac{}{\mathbf{0} \leq \mathbf{S}} \text{SUB_PURE_SPEC} \\
\\
\frac{}{\mathbf{S} \leq \mathbf{S}} \text{SUB_SPEC_SPEC}
\end{array}$$

$$\text{eff}' = \lfloor \text{eff} \rfloor$$

$$\begin{array}{c}
\frac{}{\mathbf{0} = \lfloor \mathbf{0} \rfloor} \text{PUR_PURE} \\
\\
\frac{}{\mathbf{0} = \lfloor \mathbf{1} \rfloor} \text{PUR_IMPURE} \\
\\
\frac{}{\mathbf{S} = \lfloor \mathbf{S} \rfloor} \text{PUR_SPEC}
\end{array}$$

$$\text{eff} = \text{eff}_1 \vee \text{eff}_2$$

$$\frac{}{\text{eff} = \text{eff} \vee \text{eff}} \text{JOIN_REFL}$$

$$\begin{array}{c}
\overline{\mathbf{1} = \mathbf{0} \vee \mathbf{1}} \quad \text{JOIN_PURE_IMPURE} \\
\overline{\mathbf{1} = \mathbf{1} \vee \mathbf{0}} \quad \text{JOIN_IMPURE_PURE} \\
\overline{\mathbf{S} = \mathbf{0} \vee \mathbf{S}} \quad \text{JOIN_PURE_SPEC} \\
\overline{\mathbf{S} = \mathbf{S} \vee \mathbf{0}} \quad \text{JOIN_SPEC_PURE}
\end{array}$$

$$\boxed{\text{prim}:\tau \rightarrow_{\text{eff}} \tau'}$$

$$\begin{array}{c}
\overline{\text{Add}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Int}} \quad \text{PT_ADD} \\
\overline{\text{Sub}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Int}} \quad \text{PT_SUB} \\
\overline{\text{Mul}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Int}} \quad \text{PT_MUL} \\
\overline{\text{Div}:(\text{Int}, \text{Int}) \rightarrow_1 \text{Int}} \quad \text{PT_DIV} \\
\overline{\text{And}:(\text{Bool}, \text{Bool}) \rightarrow_0 \text{Bool}} \quad \text{PT_AND} \\
\overline{\text{Or}:(\text{Bool}, \text{Bool}) \rightarrow_0 \text{Bool}} \quad \text{PT_OR} \\
\overline{\text{Not}:\text{Bool} \rightarrow_0 \text{Bool}} \quad \text{PT_NOT} \\
\overline{\text{Lt}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Bool}} \quad \text{PT_LT} \\
\overline{\text{Le}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Bool}} \quad \text{PT_LE} \\
\overline{\text{Gt}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Bool}} \quad \text{PT_GT} \\
\overline{\text{Ge}:(\text{Int}, \text{Int}) \rightarrow_0 \text{Bool}} \quad \text{PT_GE} \\
\frac{A \text{ eqtype}}{\overline{\text{Eq}[A]:(\text{A}, \text{A}) \rightarrow_0 \text{Bool}}} \quad \text{PT_EQ} \\
\overline{\text{New}[A]:\text{A} \rightarrow_1 \text{Ptr } A} \quad \text{PT_NEW} \\
\overline{\text{Del}[A]:\text{Ptr } A \rightarrow_1 ()} \quad \text{PT_DEL} \\
\overline{\text{Get}[A]:\text{Ptr } A \rightarrow_1 A} \quad \text{PT_GET} \\
\overline{\text{Set}[A]:(\text{Ptr } A, A) \rightarrow_1 ()} \quad \text{PT_SET} \\
\overline{\text{Own}[A]:\text{Ptr } A \rightarrow_{\text{S}} \text{Pred } (A)} \quad \text{PT_OWN}
\end{array}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce \Rightarrow \tau}$$

$$\begin{array}{c}
\overline{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{lit} \Rightarrow \text{Int}} \quad \text{SYN_INT} \\
\overline{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{blit} \Rightarrow \text{Bool}} \quad \text{SYN_BOOL} \\
\frac{x:\text{eff}'\tau \in \Gamma \quad \text{eff}' \leq \text{eff}_0}{\overline{\Sigma_c; \Gamma \vdash_{\text{eff}_0} x \Rightarrow \tau}} \quad \text{SYN_VAR} \\
\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce \Leftarrow \tau}{\overline{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce : \tau \Rightarrow \tau}} \quad \text{SYN_ANNOT}
\end{array}$$

$$\begin{array}{c}
\text{prim}:\tau \rightarrow_{\text{eff}'} \tau' \\
\text{eff}' \leq \text{eff}_0 \\
\text{eff}'' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}''} ce \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{prim } ce \Rightarrow \tau' \quad \text{SYN_APP}
\end{array}$$

$$\begin{array}{c}
f:\tau \rightarrow_{\text{eff}'} \tau' \in \Sigma_c \\
\text{eff}' \leq \text{eff}_0 \\
\text{eff}'' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}''} ce \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} f \text{ } ce \Rightarrow \tau' \quad \text{SYN_CALL}
\end{array}$$

$$\begin{array}{c}
A \text{ eqtype} \\
\text{eff}' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}'} ce_1 \Rightarrow A \\
\Sigma_c; \Gamma \vdash_{\text{eff}'} ce_2 \Leftarrow A \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_1 \doteq ce_2 \Rightarrow \text{Bool} \quad \text{SYN_EQ}
\end{array}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce \Leftarrow \tau}$$

$$\begin{array}{c}
\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_1 \Rightarrow \tau \\
\epsilon = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma, x:\epsilon \tau \vdash_{\text{eff}_0} ce_2 \Leftarrow \tau' \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{let } x = ce_1; ce_2 \Leftarrow \tau' \quad \text{CHK_LET}
\end{array}$$

$$\begin{array}{c}
\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_1 \Leftarrow \tau_1 \quad \dots \quad \Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_n \Leftarrow \tau_n \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} (ce_1, \dots, ce_n) \Leftarrow (\tau_1, \dots, \tau_n) \quad \text{CHK_TUPLE}
\end{array}$$

$$\begin{array}{c}
\epsilon = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\epsilon} ce_1 \Rightarrow (\tau_1, \dots, \tau_n) \\
\Gamma' = x_1 : \tau_1[\epsilon], \dots, x_n : \tau_n[\epsilon] \\
\Sigma_c; \Gamma, \Gamma' \vdash_{\text{eff}_0} ce_2 \Leftarrow \tau' \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{let } (x_1, \dots, x_n) = ce_1; ce_2 \Leftarrow \tau' \quad \text{CHK_LETTUPLE}
\end{array}$$

$$\begin{array}{c}
L : \tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \\
\text{eff}' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}'} ce \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} L \text{ } ce \Leftarrow D(\tau_1, \dots, \tau_i) \quad \text{CHK_INJECT}
\end{array}$$

$$\begin{array}{c}
\text{eff}' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}'} ce \Rightarrow D(\tau_1, \dots, \tau_i) \\
D(a_1, \dots, a_k) = \{L_1 : \tau'_1 \mid \dots \mid L_n : \tau'_n\} \in \Sigma_c \\
L_1 : \tau'_1 \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \quad \dots \quad L_n : \tau'_n \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \\
\Sigma_c; \Gamma, x_1:\text{eff}'\tau'_1 \vdash_{\text{eff}_0} ce_1 \Leftarrow \tau \quad \dots \quad \Sigma_c; \Gamma, x_n:\text{eff}'\tau'_n \vdash_{\text{eff}_0} ce_n \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{case } ce \text{ of } \{L_1 x_1 \rightarrow ce_1 \mid \dots \mid L_n x_n \rightarrow ce_n\} \Leftarrow \tau \quad \text{CHK_CASE}
\end{array}$$

$$\begin{array}{c}
\text{type } D(a, b) = \{\text{Next}:a \mid \text{Done}:b\} \in \Sigma_c \\
\Sigma_c; \Gamma \vdash_0 ce_1 \Rightarrow A \\
\Sigma_c; \Gamma, x:0A \vdash_1 ce_2 \Leftarrow D(A, B) \\
\hline
\Sigma_c; \Gamma \vdash_1 \text{iter } (x = ce_1)\{ce_2\} \Leftarrow B \quad \text{CHK_ITER}
\end{array}$$

$$\begin{array}{c}
\text{eff}' = \lfloor \text{eff}_0 \rfloor \\
\Sigma_c; \Gamma \vdash_{\text{eff}'} ce_1 \Leftarrow \text{Bool} \\
\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_2 \Leftarrow \tau \\
\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce_3 \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{if } ce_1 \text{ then } ce_2 \text{ else } ce_3 \Leftarrow \tau \quad \text{CHK_IF}
\end{array}$$

$$\begin{array}{c}
\Sigma_c; \Gamma \vdash_{\text{S}} ce \Leftarrow \tau \\
\hline
\Sigma_c; \Gamma \vdash_{\text{S}} \text{return } ce \Leftarrow \text{Pred } \tau \quad \text{CHK_RETURN}
\end{array}$$

$$\begin{array}{c}
\frac{}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \text{fail} \Leftarrow \text{Pred } \tau} \text{CHK_FAIL} \\
\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \text{Pred } \tau \quad \Sigma_c; \Gamma, x:\mathbf{S}\tau \vdash_{\mathbf{S}} ce_2 \Leftarrow \text{Pred } \tau'}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \text{take } x = ce_1; ce_2 \Leftarrow \text{Pred } \tau'} \text{CHK_TAKE} \\
\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce \Rightarrow \tau}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} ce \Leftarrow \tau} \text{CHK_SUB}
\end{array}$$

$$\boxed{\Sigma_c \vdash cprog}$$

$$\begin{array}{c}
\frac{\Sigma_c; \cdot \vdash_{\text{eff}} ce \Leftarrow \tau}{\Sigma_c \vdash \text{main} : \tau_{\text{eff}} = ce} \text{CPROG_MAIN} \\
\frac{\Sigma_c; \cdot, x:\mathbf{0}A \vdash_{\mathbf{0}} ce \Leftarrow B \quad \Sigma_c, \text{fun } f(x:A) \rightarrow_{\mathbf{0}} B = ce \vdash cprog}{\Sigma_c \vdash \text{fun } f(x:A) \rightarrow_{\mathbf{0}} B = ce \quad cprog} \text{CPROG_PUREFUN} \\
\frac{\Sigma_c, f:A \rightarrow_{\mathbf{1}} B; \cdot, x:\mathbf{0}A \vdash_{\mathbf{1}} ce \Leftarrow B \quad \Sigma_c, f:A \rightarrow_{\mathbf{1}} B \vdash cprog}{\Sigma_c \vdash \text{fun } f(x:A) \rightarrow_{\mathbf{1}} B = ce \quad cprog} \text{CPROG_IMPUREFUN} \\
\frac{\Sigma_c, f:\tau \rightarrow_{\mathbf{S}} \tau'; \cdot, x:\mathbf{S}\tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \quad \Sigma_c, \text{fun } f(x:\tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash cprog}{\Sigma_c \vdash \text{fun } f(x:\tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad cprog} \text{CPROG_SPECFUN} \\
\frac{\Sigma_c \vdash \text{sort } D(\bar{a}) = \{\overline{L:\tau}\} \text{ ok} \quad \Sigma_c, \text{sort } D(\bar{a}) = \{\overline{L:\tau}\} \vdash cprog}{\Sigma_c \vdash \text{sort } D(a_1, \dots, a_i) = \{L_1:\tau_1 \mid \dots \mid L_n:\tau_n\} cprog} \text{CPROG_SORTDECL} \\
\frac{\Sigma_c \vdash \text{type } D(\bar{a}) = \{\overline{L:A}\} \text{ ok} \quad \Sigma_c, \text{type } D(\bar{a}) = \{\overline{L:A}\} \vdash cprog}{\Sigma_c \vdash \text{type } D(a_1, \dots, a_i) = \{L_1:A_1 \mid \dots \mid L_n:A_n\} cprog} \text{CPROG_TYPEDECL}
\end{array}$$

$$\boxed{L:\tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c}$$

$$\begin{array}{c}
\frac{\text{sort } D(a_1, \dots, a_i) \text{ in } \Sigma_c \quad L:\tau' \in \text{ctors}(D) \in \Sigma_c \quad \tau = [\cdot, \tau_1/a_1, \dots, \tau_i/a_i]\tau'}{L:\tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c} \text{CTOR_SORT} \\
\frac{\text{type } D(a_1, \dots, a_i) \text{ in } \Sigma_c \quad L:A' \in \text{tyctors}(D) \in \Sigma_c \quad A = [\cdot, A_1/a_1, \dots, A_i/a_i]A'}{L:A \text{ in } D(A_1, \dots, A_i) \text{ in } \Sigma_c} \text{CTOR_TYPE}
\end{array}$$

$$\boxed{D(a_1, \dots, a_k) = \{L_1:\tau_1 \mid \dots \mid L_n:\tau_n\} \in \Sigma_c}$$

$$\begin{array}{c}
\frac{\text{sort } D(a_1, \dots, a_k) = \{L_1:\tau_1 \mid \dots \mid L_n:\tau_n\} \text{ in } \Sigma_c}{D(a_1, \dots, a_k) = \{L_1:\tau_1 \mid \dots \mid L_n:\tau_n\} \in \Sigma_c} \text{DL_SORT} \\
\frac{\text{type } D(a_1, \dots, a_k) = \{L_1:A_1 \mid \dots \mid L_n:A_n\} \text{ in } \Sigma_c}{D(a_1, \dots, a_k) = \{L_1:A_1 \mid \dots \mid L_n:A_n\} \in \Sigma_c} \text{DL_TYPE}
\end{array}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}$$

$$\begin{array}{c}
\frac{x:\text{eff}'\tau \in \Gamma \quad \text{eff}' \leq \text{eff}_0}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} x \Rightarrow \tau \rightsquigarrow x} \text{SURFSYN_VAR} \\
\frac{}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{lit} \Rightarrow \text{Int} \rightsquigarrow \text{lit}} \text{SURFSYN_INT} \\
\frac{}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{blit} \Rightarrow \text{Bool} \rightsquigarrow \text{blit}} \text{SURFSYN_BOOL}
\end{array}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se : \tau \Rightarrow \tau \rightsquigarrow ce : \tau} \text{SURFSYN_ANNOT}$$

$$\frac{\begin{array}{l} \text{prim} : \tau \rightarrow_{\text{eff}'} \tau' \\ \text{eff}' \leq \text{eff}_0 \\ \text{eff}'' = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{prim } se \Rightarrow \tau' \rightsquigarrow \text{prim } ce} \text{SURFSYN_APP}$$

$$\frac{\begin{array}{l} f : \tau \rightarrow_{\text{eff}'} \tau' \in \Sigma_c \\ \text{eff}' \leq \text{eff}_0 \\ \text{eff}'' = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}''} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} f \text{ } se \Rightarrow \tau' \rightsquigarrow f \text{ } ce} \text{SURFSYN_CALL}$$

$$\frac{\begin{array}{l} A \text{ eqtype} \\ \text{eff}' = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}'} se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\text{eff}'} se_2 \Leftarrow A \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se_1 \doteq se_2 \Rightarrow \text{Bool} \rightsquigarrow ce_1 \doteq ce_2} \text{SURFSYN_EQ}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se \Leftarrow \tau \rightsquigarrow ce}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_{\text{eff}_0} se_1 \Rightarrow \tau \rightsquigarrow ce_1 \\ \epsilon = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}_0} y, \cdot; \text{pat} : \tau \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_{\epsilon} \tau' \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{let } \text{pat} = se_1; se_2 \Leftarrow \tau' \rightsquigarrow \text{let } y = ce_1; ce_2} \text{SURFCHK_LET}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se_1 \Leftarrow \tau_1 \rightsquigarrow ce_1 \quad \dots \quad \Sigma_c; \Gamma \vdash_{\text{eff}_0} se_n \Leftarrow \tau_n \rightsquigarrow ce_n}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} (se_1, \dots, se_n) \Leftarrow (\tau_1, \dots, \tau_n) \rightsquigarrow (ce_1, \dots, ce_n)} \text{SURFCHK_TUPLE}$$

$$\frac{\begin{array}{l} L : \tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \\ \text{eff}' = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}'} se \Leftarrow \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} L \text{ } se \Leftarrow D(\tau_1, \dots, \tau_i) \rightsquigarrow L \text{ } ce} \text{SURFCHK_INJECT}$$

$$\frac{\begin{array}{l} \epsilon = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\epsilon} se \Rightarrow \tau \rightsquigarrow ce' \\ M = \text{pat} : \tau, \cdot \rightarrow [\cdot][[\cdot]] se \\ \Sigma_c; \Gamma \vdash_{\text{eff}_0} y, \cdot; M \Leftarrow_{\epsilon} \tau' \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{case } se \text{ of } \{ \text{pat}_1 \rightarrow se_1 \mid \dots \mid \text{pat}_n \rightarrow se_n \} \Leftarrow \tau' \rightsquigarrow \text{let } y = ce'; ce} \text{SURFCHK_CASE}$$

$$\frac{\begin{array}{l} \Sigma_c; \Gamma \vdash_0 se_1 \Rightarrow A \rightsquigarrow ce_1 \\ \text{type } D(a, b) = \{ \text{Next} : a \mid \text{Done} : b \} \in \Sigma_c \\ \Sigma_c; \Gamma \vdash_1 z, \cdot; \text{pat} : A, \cdot \rightarrow [\cdot][[\cdot]] se_2 \Leftarrow_0 D(A, B) \rightsquigarrow ce_2 \end{array}}{\Sigma_c; \Gamma \vdash_1 \text{iter } (\text{pat} = se_1) \{ se_2 \} \Leftarrow B \rightsquigarrow \text{iter } (z = ce_1) \{ ce_2 \}} \text{SURFCHK_ITER}$$

$$\frac{\begin{array}{l} \text{eff}' = \lfloor \text{eff}_0 \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}'} se_1 \Leftarrow \text{Bool} \rightsquigarrow ce_1 \\ \Sigma_c; \Gamma \vdash_{\text{eff}_0} se_2 \Leftarrow \tau \rightsquigarrow ce_2 \\ \Sigma_c; \Gamma \vdash_{\text{eff}_0} se_3 \Leftarrow \tau \rightsquigarrow ce_3 \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \text{if } se_1 \text{ then } se_2 \text{ else } se_3 \Leftarrow \tau \rightsquigarrow \text{if } ce_1 \text{ then } ce_2 \text{ else } ce_3} \text{SURFCHK_IF}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{S}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\text{S}} \text{return } se \Leftarrow \text{Pred } \tau \rightsquigarrow \text{return } ce} \text{SURFCHK_RETURN}$$

$$\frac{}{\Sigma_c; \Gamma \vdash_{\text{S}} \text{fail} \Leftarrow \text{Pred } \tau \rightsquigarrow \text{fail}} \text{SURFCHK_FAIL}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\mathbf{S}} se_1 \Rightarrow \text{Pred } \tau \rightsquigarrow ce_1 \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} y, \cdot; \text{pat} : \tau \rightarrow [\cdot][\cdot] se_2 \Leftarrow_{\mathbf{S}} \text{Pred } \tau' \rightsquigarrow ce_2}{\Sigma_c; \Gamma \vdash_{\mathbf{S}} \text{take pat} = se_1; se_2 \Leftarrow \text{Pred } \tau' \rightsquigarrow \text{take } y = ce_1; ce_2} \quad \text{SURFCHK_TAKE}$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se \Rightarrow \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} se \Leftarrow \tau \rightsquigarrow ce} \quad \text{SURFCHK_SUB}$$

$$\boxed{\Sigma_s \vdash \text{prog} \rightsquigarrow \Sigma_c \vdash \text{cprog}}$$

$$\frac{\Sigma_c; \cdot \vdash_{\text{eff}} se \Leftarrow \tau \rightsquigarrow ce}{\Sigma_s \vdash \text{main} : \tau_{\text{eff}} = se \rightsquigarrow \Sigma_c \vdash \text{main} : \tau_{\text{eff}} = ce} \quad \text{PROG_MAIN}$$

$$\frac{\begin{array}{l} M = \overline{\text{pat}:A, \cdot \rightarrow [\cdot][\cdot] se} \\ \Sigma_c; \cdot \vdash_0 y, \cdot; M \Leftarrow_0 B \rightsquigarrow ce \\ \Sigma_s, \text{fun } f : A \rightarrow_0 B = \{\overline{\text{pat}} \rightarrow \overline{\text{se}}\} \vdash \text{prog} \\ \rightsquigarrow \Sigma_c, \text{fun } f(y : A) \rightarrow_0 B = ce \vdash \text{cprog} \end{array}}{\Sigma_s \vdash \text{fun } f : A \rightarrow_0 B = \{\overline{\text{pat}} \rightarrow \overline{\text{se}}\} \text{ prog} \rightsquigarrow \Sigma_c \vdash \text{fun } f(y : A) \rightarrow_0 B = ce \text{ cprog}} \quad \text{PROG_PUREFUN}$$

$$\frac{\begin{array}{l} M = \overline{\text{pat}:A, \cdot \rightarrow [\cdot][\cdot] se} \\ \Sigma_c, f : A \rightarrow_1 B; \cdot \vdash_1 y, \cdot; M \Leftarrow_0 B \rightsquigarrow ce \\ \Sigma_s, f : A \rightarrow_1 B \vdash \text{prog} \\ \rightsquigarrow \Sigma_c, f : A \rightarrow_1 B \vdash \text{cprog} \end{array}}{\Sigma_s \vdash \text{fun } f : A \rightarrow_1 B = \{\overline{\text{pat}} \rightarrow \overline{\text{se}}\} \text{ prog} \rightsquigarrow \Sigma_c \vdash \text{fun } f(y : A) \rightarrow_1 B = ce \text{ cprog}} \quad \text{PROG_IMPUREFUN}$$

$$\frac{\begin{array}{l} M = \overline{\text{pat}:\tau, \cdot \rightarrow [\cdot][\cdot] se} \\ \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot \vdash_{\mathbf{S}} y, \cdot; M \Leftarrow_{\mathbf{S}} \tau' \rightsquigarrow ce \\ \Sigma_s, \text{fun } f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\text{pat}} \rightarrow \overline{\text{se}}\} \vdash \text{prog} \\ \rightsquigarrow \Sigma_c, \text{fun } f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash \text{cprog} \end{array}}{\Sigma_s \vdash \text{fun } f : \tau \rightarrow_{\mathbf{S}} \tau' = \{\overline{\text{pat}} \rightarrow \overline{\text{se}}\} \text{ prog} \rightsquigarrow \Sigma_c \vdash \text{fun } f(y : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \text{ cprog}} \quad \text{PROG_SPECFUN}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \text{sort } D(\bar{a}) = \{\overline{L : \tau}\} \text{ ok} \\ \Sigma_s, \text{sort } D(\bar{a}) = \{\overline{L : \tau}\} \vdash \text{prog} \\ \rightsquigarrow \Sigma_c, \text{sort } D(\bar{a}) = \{\overline{L : \tau}\} \vdash \text{cprog} \end{array}}{\Sigma_s \vdash \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ prog} \rightsquigarrow \Sigma_c \vdash \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{ cprog}} \quad \text{PROG_SORTDECL}$$

$$\frac{\begin{array}{l} \Sigma_c \vdash \text{type } D(\bar{a}) = \{\overline{L : A}\} \text{ ok} \\ \Sigma_s, \text{type } D(\bar{a}) = \{\overline{L : A}\} \vdash \text{prog} \\ \rightsquigarrow \Sigma_c, \text{type } D(\bar{a}) = \{\overline{L : A}\} \vdash \text{cprog} \end{array}}{\Sigma_s \vdash \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{ prog} \rightsquigarrow \Sigma_c \vdash \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{ cprog}} \quad \text{PROG_TYPEDECL}$$

$$\boxed{M \text{ has_con}}$$

$$\overline{L \text{ pat} : \tau, \beta \rightarrow [\Gamma][\mathcal{E}] se \mid M \text{ has_con}} \quad \text{HC_HERE}$$

$$\frac{M \text{ has_con}}{\beta \rightarrow [\Gamma][\mathcal{E}] se \mid M \text{ has_con}} \quad \text{HC_THERE}$$

$$\boxed{M \text{ has_tup}}$$

$$\overline{(\text{pat}_1, \dots, \text{pat}_n) : \tau, \beta \rightarrow [\Gamma][\mathcal{E}] se \mid M \text{ has_tup}} \quad \text{HT_HERE}$$

$$\frac{M \text{ has_tup}}{\beta \rightarrow [\Gamma][\mathcal{E}]se \mid M \text{ has_tup}} \quad \text{HT_THERE}$$

$$\boxed{M/\text{var} \xrightarrow{y}_{\text{eff}} M'}$$

$$\frac{}{\cdot/\text{var} \xrightarrow{y}_{\text{eff}} \cdot} \quad \text{SV_EMPTY}$$

$$\frac{M/\text{var} \xrightarrow{y}_{\text{eff}} M'}{x : \tau, \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/\text{var} \xrightarrow{y}_{\text{eff}} \beta \rightarrow [\Gamma, x:\text{eff}\tau][\text{let } x = y; \mathcal{E}]se \mid M'} \quad \text{SV_CONS}$$

$$\boxed{M/L:\tau \xrightarrow{y}_{\text{eff}} M'}$$

$$\frac{}{\cdot/L:\tau \xrightarrow{y}_{\text{eff}} \cdot} \quad \text{SC_EMPTY}$$

$$\frac{M/L:\tau \xrightarrow{y}_{\text{eff}} M'}{L \text{ pat} : \tau', \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/L:\tau \xrightarrow{y}_{\text{eff}} \text{pat} : \tau, \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M'} \quad \text{SC_MATCH}$$

$$\frac{L \neq L' \quad M/L:\tau \xrightarrow{y}_{\text{eff}} M'}{L' \text{ pat} : \tau', \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/L:\tau \xrightarrow{y}_{\text{eff}} M'} \quad \text{SC_SKIP}$$

$$\frac{M/L:\tau \xrightarrow{y}_{\text{eff}} M'}{x : \tau', \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/L:\tau \xrightarrow{y}_{\text{eff}} z : \tau, \beta \rightarrow [\Gamma, x:\text{eff}\tau'][\text{let } x = y; \mathcal{E}]se \mid M'} \quad \text{SC_VAR}$$

$$\boxed{M/\text{tup} \xrightarrow{y}_{\text{eff}} M'}$$

$$\frac{}{\cdot/\text{tup} \xrightarrow{y}_{\text{eff}} \cdot} \quad \text{ET_EMPTY}$$

$$\frac{M/\text{tup} \xrightarrow{y}_{\text{eff}} M'}{(\text{pat}_1, \dots, \text{pat}_k) : (\tau_1, \dots, \tau_k), \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/\text{tup} \xrightarrow{y}_{\text{eff}} \text{pat}_1 : \tau_1, \dots, \text{pat}_k : \tau_k, \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M'} \quad \text{ET_TUPLE}$$

$$\frac{M/\text{tup} \xrightarrow{y}_{\text{eff}} M'}{x : (\tau_1, \dots, \tau_k), \beta \rightarrow [\Gamma][\mathcal{E}]se \mid M/\text{tup} \xrightarrow{y}_{\text{eff}} z_1 : \tau_1, \dots, z_k : \tau_k, \beta \rightarrow [\Gamma, x:\text{eff}(\tau_1, \dots, \tau_k)][\text{let } x = y; \mathcal{E}]se \mid M'} \quad \text{ET_VAR}$$

$$\boxed{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \bar{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow ce}$$

$$\frac{\Sigma_c; \Gamma, \Gamma' \vdash_{\text{eff}_0} se \Leftarrow \tau \rightsquigarrow ce' \quad ce = \mathcal{E}[ce']}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} \cdot; \cdot \rightarrow [\Gamma'][\mathcal{E}]se \mid M \Leftarrow_{\epsilon} \tau \rightsquigarrow ce} \quad \text{COV_DONE}$$

$$\frac{\begin{array}{c} M/\text{var} \xrightarrow{y}_{\epsilon} M' \\ \Sigma_c; \Gamma \vdash_{\text{eff}_0} \bar{y}; M' \Leftarrow_{\epsilon} \tau \rightsquigarrow ce \end{array}}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} y, \bar{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow ce} \quad \text{COV_VAR}$$

$M \text{ has_con}$

$$D(a_1, \dots, a_k) = \{L_1 : \tau'_1 \mid \dots \mid L_n : \tau'_n\} \in \Sigma_c$$

$$L_1 : \tau''_1 \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \quad \dots \quad L_n : \tau''_n \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c$$

$$M/L_1:\tau''_1 \xrightarrow{y}_{\epsilon} M_1 \quad \dots \quad M/L_n:\tau''_n \xrightarrow{y}_{\epsilon} M_n$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} x_1, \bar{y}; M_1 \Leftarrow_{\epsilon} \tau \rightsquigarrow ce_1 \quad \dots \quad \Sigma_c; \Gamma \vdash_{\text{eff}_0} x_n, \bar{y}; M_n \Leftarrow_{\epsilon} \tau \rightsquigarrow ce_n}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} y, \bar{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow \text{case } y \text{ of } \{L_1 x_1 \rightarrow ce_1 \mid \dots \mid L_n x_n \rightarrow ce_n\}} \quad \text{COV_CON}$$

$M \text{ has_tup}$

$$M/\text{tup} \xrightarrow{y}_{\epsilon} M'$$

$$\frac{\Sigma_c; \Gamma \vdash_{\text{eff}_0} y_1, \dots, y_n, \bar{y}; M' \Leftarrow_{\epsilon} \tau \rightsquigarrow ce}{\Sigma_c; \Gamma \vdash_{\text{eff}_0} y, \bar{y}; M \Leftarrow_{\epsilon} \tau \rightsquigarrow \text{let } (y_1, \dots, y_n) = y; ce} \quad \text{COV_TUP}$$

$$\boxed{\mathcal{E}[ce] \rightsquigarrow ce'}$$

$$\frac{}{[\cdot][ce] \rightsquigarrow ce} \text{ FILL_HOLE}$$

$$\frac{\mathcal{E}[ce] \rightsquigarrow ce'}{\text{let } x = ce''; \mathcal{E}[ce] \rightsquigarrow \text{let } x = ce''; ce'} \text{ FILL_LET}$$

$$\boxed{\Sigma_r \vdash \Delta \text{ wf}}$$

$$\frac{}{\Sigma_r \vdash \cdot \text{ wf}} \text{ RW_F_EMPTY}$$

$$\frac{\Sigma_r \vdash \Delta \text{ wf} \quad \text{eff} \leq \mathbf{S}}{\Sigma_r \vdash \Delta, x : \tau[\text{eff}] \text{ wf}} \text{ RW_F_EFF}$$

$$\frac{\Sigma_r \vdash \Delta \text{ wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \text{Bool}}{\Sigma_r \vdash \Delta, x : ce[\log] \text{ wf}} \text{ RW_F_LOG}$$

$$\frac{\Sigma_r \vdash \Delta \text{ wf} \quad \Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \text{Pred } \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce' \Leftarrow \tau}{\Sigma_r \vdash \Delta, x : ce@ce'[\text{res}(u)] \text{ wf}} \text{ RW_F_RES}$$

$$\boxed{\Gamma = |\Delta|}$$

$$\frac{}{\cdot = |\cdot|} \text{ ERASE_EMPTY}$$

$$\frac{\Gamma = |\Delta|}{\Gamma, x :_{\text{eff}} \tau = |\Delta, x : \tau[\text{eff}]|} \text{ ERASE_EFF}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x : ce[\log]|} \text{ ERASE_LOG}$$

$$\frac{\Gamma = |\Delta|}{\Gamma = |\Delta, x : ce@ce'[\text{res}(u)]|} \text{ ERASE_RES}$$

$$\boxed{\Delta' = \text{affinize}(\Delta)}$$

$$\frac{}{\cdot = \text{affinize}(\cdot)} \text{ AFF_EMPTY}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Delta', x : \tau[\text{eff}] = \text{affinize}(\Delta, x : \tau[\text{eff}])} \text{ AFF_EFF}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Delta', x : ce[\log] = \text{affinize}(\Delta, x : ce[\log])} \text{ AFF_LOG}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Delta', x : ce@ce'[\text{res}(0)] = \text{affinize}(\Delta, x : ce@ce'[\text{res}(0)])} \text{ AFF_RES_USED}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Delta', x : ce@ce'[\text{res}(?)] = \text{affinize}(\Delta, x : ce@ce'[\text{res}(1)])} \text{ AFF_RES_AVAIL}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Delta', x : ce@ce'[\text{res}(?)] = \text{affinize}(\Delta, x : ce@ce'[\text{res}(?)])} \text{ AFF_RES_OPT}$$

$$\boxed{\text{zero}(\Delta)}$$

$$\frac{}{\text{zero}(\cdot)} \text{ ZERO_EMPTY}$$

$$\frac{\text{zero}(\Delta)}{\text{zero}(\Delta, x : \tau[\text{eff}])} \text{ ZERO_EFF}$$

$$\frac{\text{zero}(\Delta)}{\text{zero}(\Delta, x : ce[\log])} \text{ ZERO_LOG}$$

$$\frac{\text{zero}(\Delta) \quad u \in \{0, ?\}}{\text{zero}(\Delta, x : ce @ ce' [\text{res}(u)])} \quad \text{ZERO_RES}$$

$$\boxed{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}$$

$$\begin{array}{c} \overline{\mathcal{C} = \cdot \Rightarrow \mathcal{C}} \quad \text{CLOSE_EMPTY} \\ \frac{\mathcal{C} = \Delta \Rightarrow \forall x : \tau. \mathcal{C}'}{\mathcal{C} = \Delta, x : \tau[\text{eff}] \Rightarrow \mathcal{C}'} \quad \text{CLOSE_EFF} \\ \frac{\mathcal{C} = \Delta \Rightarrow ce \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x : ce[\text{log}] \Rightarrow \mathcal{C}'} \quad \text{CLOSE_LOG} \\ \frac{\mathcal{C} = \Delta \Rightarrow \mathcal{C}'}{\mathcal{C} = \Delta, x : ce @ ce' [\text{res}(u)] \Rightarrow \mathcal{C}'} \quad \text{CLOSE_RES} \end{array}$$

$$\boxed{\Sigma_r; \Gamma \vdash \mathcal{C} \text{ wf}}$$

$$\begin{array}{c} \overline{\Sigma_r; \Gamma \vdash \top \text{ wf}} \quad \text{CTWF_TOP} \\ \frac{\Sigma_r; \Gamma \vdash \mathcal{C}_1 \text{ wf} \quad \Sigma_r; \Gamma \vdash \mathcal{C}_2 \text{ wf}}{\Sigma_r; \Gamma \vdash \mathcal{C}_1 \wedge \mathcal{C}_2 \text{ wf}} \quad \text{CTWF_AND} \\ \frac{\Sigma_r; \Gamma, x : \mathbf{s}\tau \vdash \mathcal{C} \text{ wf}}{\Sigma_r; \Gamma \vdash \forall x : \tau. \mathcal{C} \text{ wf}} \quad \text{CTWF_FORALL} \\ \frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce \Leftarrow \text{Bool} \quad \Sigma_r; \Gamma \vdash \mathcal{C} \text{ wf}}{\Sigma_r; \Gamma \vdash ce \Rightarrow \mathcal{C} \text{ wf}} \quad \text{CTWF_IMPL} \\ \frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce \Leftarrow \text{Bool}}{\Sigma_r; \Gamma \vdash ce \text{ wf}} \quad \text{CTWF_ATOM} \\ \overline{\Sigma_r; \Gamma \vdash \perp \text{ wf}} \quad \text{CTWF_BOT} \end{array}$$

$$\boxed{\Sigma_r; \Gamma \vdash_{\text{eff}} Pf \text{ wf}}$$

$$\begin{array}{c} \overline{\Sigma_r; \Gamma \vdash_{\text{eff}} \cdot \text{ wf}} \quad \text{PFWF_NIL} \\ \frac{\text{eff}' \in \{\mathbf{S}, \lfloor \text{eff} \rfloor\} \quad \Sigma_r; \Gamma, x : \text{eff}'\tau \vdash_{\text{eff}} Pf \text{ wf}}{\Sigma_r; \Gamma \vdash_{\text{eff}} x : \text{eff}'\tau, Pf \text{ wf}} \quad \text{PFWF_COMP} \\ \frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce \Leftarrow \text{Bool} \quad \Sigma_r; \Gamma \vdash_{\text{eff}} Pf \text{ wf}}{\Sigma_r; \Gamma \vdash_{\text{eff}} ce [\text{log}], Pf \text{ wf}} \quad \text{PFWF_LOG} \\ \frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce' \Rightarrow \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce \Leftarrow \text{Pred } \tau \quad \Sigma_r; \Gamma \vdash_{\text{eff}} Pf \text{ wf}}{\Sigma_r; \Gamma \vdash_{\text{eff}} ce @ ce' [\text{res}], Pf \text{ wf}} \quad \text{PFWF_RES} \\ \frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{s}} ce \Rightarrow \text{Pred } \tau \quad \Sigma_r; \Gamma, y : \mathbf{s}\tau \vdash_{\text{eff}} Pf \text{ wf}}{\Sigma_r; \Gamma \vdash_{\text{eff}} (y).ce [\text{res}], Pf \text{ wf}} \quad \text{PFWF_DEPRES} \end{array}$$

$$\boxed{\Sigma_r \vdash RF \text{ wf}}$$

$$\frac{\Sigma_r; \cdot \vdash_{\text{eff}} Pf_1 \text{ wf} \quad \Gamma = \text{Bind}(\cdot; Pf_1) \quad \Sigma_r; \Gamma \vdash_{\text{eff}} Pf_2 \text{ wf}}{\Sigma_r \vdash Pf_1 \multimap Pf_2 [\text{eff}] \text{ wf}} \quad \text{RWWF_RFUN}$$

$$\boxed{\Sigma_r \vdash f : RF}$$

$$\frac{f : RF \in \Sigma_r}{\Sigma_r \vdash f : RF} \quad \text{RSLK_RFUN}$$

$$\frac{f:\tau \rightarrow_{\text{eff}} \tau' \in \Sigma_r}{\Sigma_r \vdash f:x:\text{eff}\tau, \cdot \multimap y:\text{eff}\tau', \cdot [\text{eff}]} \text{RSLK_SIG}$$

$$\frac{f(\mathbf{x} : \tau) \rightarrow_{\text{eff}} \tau' = ce \in \Sigma_r}{\Sigma_r \vdash f:x:\text{eff}\tau, \cdot \multimap y:\text{eff}\tau', \cdot [\text{eff}]} \text{RSLK_DEF}$$

$$\overline{A} = \{Pf\}$$

$$\overline{[] = \{\cdot\}} \text{PFTYS_NIL}$$

$$\frac{\overline{A} = \{Pf\}}{A::\overline{A} = \{x:\mathbf{0}A, Pf\}} \text{PFTYS_PURE}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{x:\mathbf{S}\tau, Pf\}} \text{PFTYS_SPEC}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce [\log], Pf\}} \text{PFTYS_LOG}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{ce@ce' [\text{res}], Pf\}} \text{PFTYS_RES}$$

$$\frac{\overline{A} = \{Pf\}}{\overline{A} = \{(y).ce [\text{res}], Pf\}} \text{PFTYS_DEPRES}$$

$$\Sigma_c = \text{Comp}(\Sigma_r)$$

$$\overline{\cdot = \text{Comp}(\cdot)} \text{RSErase_EMPTY}$$

$$\frac{\Sigma_c = \text{Comp}(\Sigma_r)}{\Sigma_c, f : F = \text{Comp}(\Sigma_r, f : F)} \text{RSErase_EXT}$$

$$\frac{\Sigma_c = \text{Comp}(\Sigma_r)}{\Sigma_c, \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = ce = \text{Comp}(\Sigma_r, \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = ce)} \text{RSErase_FUNDEF}$$

$$\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \overline{A_1} = \{Pf_1\} \quad A = \text{Prod}(\overline{A_1}) \quad \overline{A_2} = \{Pf_2\} \quad B = \text{Prod}(\overline{A_2})}{\Sigma_c, f : A \rightarrow_{\text{eff}} B = \text{Comp}(\Sigma_r, f : Pf_1 \multimap Pf_2 [\text{eff}])} \text{RSErase_RFUN}$$

$$\frac{\Sigma_c = \text{Comp}(\Sigma_r)}{\Sigma_c, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\} = \text{Comp}(\Sigma_r, \text{sort } D(\overline{a}) = \{\overline{L : \tau}\})} \text{RSErase_SORT}$$

$$\frac{\Sigma_c = \text{Comp}(\Sigma_r)}{\Sigma_c, \text{type } D(\overline{a}) = \{\overline{L : A}\} = \text{Comp}(\Sigma_r, \text{type } D(\overline{a}) = \{\overline{L : A}\})} \text{RSErase_TYPE}$$

$$\vdash \Sigma_r \text{ wf}$$

$$\overline{\vdash \cdot \text{ wf}} \text{RSWF_EMPTY}$$

$$\frac{\vdash \Sigma_r \text{ wf}}{\vdash \Sigma_r, f : F \text{ wf}} \text{RSWF_EXT}$$

$$\frac{\vdash \Sigma_r \text{ wf} \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma, x:\text{eff}\tau \vdash_{\text{eff}} ce \Leftarrow \tau'}{\vdash \Sigma_r, \text{fun } f(x : \tau) \rightarrow_{\text{eff}} \tau' = ce \text{ wf}} \text{RSWF_FUNDEF}$$

$$\frac{\vdash \Sigma_r \text{ wf} \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_r \vdash RF \text{ wf}}{\vdash \Sigma_r, f : RF \text{ wf}} \text{RSWF_RFUN}$$

$$\Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}$$

$$\frac{x:ce [\log] \in \Delta}{\Sigma_r; \Delta \vdash x \Rightarrow ce \dashv \Delta \rightsquigarrow \top} \text{LSYN_VAR}$$

$$\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \rightarrow_{\text{eff}} \tau' = ce' \in \Sigma_c \quad \text{eff} \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \end{array}}{\Sigma_r; \Delta \vdash \text{unfold } f(ce) \Rightarrow (f \text{ } ce) \doteq [\text{id}(\Gamma), (ce : \tau)/x] ce' \dashv \Delta \rightsquigarrow \top} \text{LSYN_UNFOLD}$$

$$\frac{\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash lpf : ce \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{LSYN_ANNOT}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow \text{return } ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \text{open-ret } rpf \Rightarrow ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{LSYN_OPENRET}$$

$$\boxed{\Sigma_r; \Delta \vdash lpf \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash \text{auto} \Leftarrow ce \dashv \Delta \rightsquigarrow ce} \text{LCHK_AUTO}$$

$$\frac{\Sigma_r; \Delta \vdash lpf \Rightarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash lpf \Leftarrow ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge (ce \Rightarrow (ce'))} \text{LCHK_SUB}$$

$$\boxed{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : \tau[\text{eff}] \vdash x : ce@ce' \dashv \Delta', y : \tau[\text{eff}]} \text{RLK_THERE_EFF}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : ce''[\log] \vdash x : ce@ce' \dashv \Delta', y : ce''[\log]} \text{RLK_THERE_LOG}$$

$$\frac{x \neq y \quad \Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta, y : ce''@ce'''[\text{res}(u)] \vdash x : ce@ce' \dashv \Delta', y : ce''@ce'''[\text{res}(u)]} \text{RLK_THERE_RES}$$

$$\frac{u \in \{1, ?\}}{\Sigma_r; \Delta, x : ce@ce'[\text{res}(u)] \vdash x : ce@ce' \dashv \Delta, x : ce@ce'[\text{res}(0)]} \text{RLK_HERE}$$

$$\boxed{\Sigma_r; \Delta \vdash rpf \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\Sigma_r; \Delta \vdash x : ce@ce' \dashv \Delta'}{\Sigma_r; \Delta \vdash x \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \top} \text{RSYN_VAR}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf : ce@ce' \Rightarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RSYN_ANNOT}$$

$$\boxed{\Sigma_r; \Delta \vdash rpf \Leftarrow ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\Sigma_r; \Delta \vdash lpf \Leftarrow ce_1 \doteq ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \text{return } lpf \Leftarrow \text{return } ce_1@ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RCHK_RETURN}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta_1| \\ \Sigma_r; \Delta_1 \vdash rpf_1 \Rightarrow ce_1@ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash rpf_2 \Leftarrow [\text{id}(\Gamma), ce/x] ce_2@ce_3 \dashv \Delta_3 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_1 \vdash \text{take}(rpf_1, rpf_2) \Leftarrow \text{take } x = ce_1; ce_2@ce_3 \dashv \Delta_3 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \text{RCHK_TAKE}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash lpf \Leftarrow \text{false} \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Delta'' = \text{affinize}(\Delta') \end{array}}{\Sigma_r; \Delta \vdash \text{fail}[lpf] \Leftarrow \text{fail}@ce \dashv \Delta'' \rightsquigarrow \mathcal{C}} \text{RCHK_FAIL}$$

$$\begin{array}{l}
\Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \tau \\
\Sigma_r; \Delta_0 \vdash_{\mathbf{S}} cpat : \tau \dashv \Delta_1 \rightsquigarrow ce \\
\Sigma_r; \Delta_1 \vdash lpat : ce \doteq ce_1 \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\
\Sigma_r; \Delta_2 \vdash rpf \Leftarrow [\text{id}(\Gamma), (ce : \tau)/x] ce_2 @ ce_3 \dashv \Delta_3, \Delta'' \rightsquigarrow \mathcal{C}_2 \\
\text{len}(\Delta_0) = \text{len}(\Delta_3) \\
\text{zero}(\Delta'') \\
\mathcal{C} = \Delta'' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2)
\end{array}$$

$$\frac{}{\Sigma_r; \Delta_0 \vdash \text{let } [lpat] cpat; rpf \Leftarrow \text{let } x = ce_1; ce_2 @ ce_3 \dashv \Delta_3 \rightsquigarrow \mathcal{C}} \text{RCHK_LET}$$

$$\begin{array}{l}
\Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \\
\Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow D(\tau_1, \dots, \tau_i) \\
L : \tau \in \text{ctors}(D) \in \Sigma_c \\
ce'' = \text{branch}(L, x, ce''') \\
\Sigma_r; \Delta_0 \vdash_{\mathbf{S}} cpat : \tau \dashv \Delta_1 \rightsquigarrow ce_0 \\
\Sigma_r; \Delta_1 \vdash lpat : (L ce_0) \doteq ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\
\Sigma_r; \Delta_2 \vdash rpf \Leftarrow [\text{id}(\Gamma), (ce_0 : \tau)/x] ce'' @ ce' \dashv \Delta_3, \Delta'' \rightsquigarrow \mathcal{C}_2 \\
\text{len}(\Delta_0) = \text{len}(\Delta_3) \\
\text{zero}(\Delta'') \\
\mathcal{C} = \Delta'' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2)
\end{array}$$

$$\frac{}{\Sigma_r; \Delta_0 \vdash \text{case } [lpat] L cpat; rpf \Leftarrow ce''' @ ce' \dashv \Delta_3 \rightsquigarrow (\text{is } ce L) \wedge \mathcal{C}} \text{RCHK_CASE}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce_2 @ ce_4 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \text{iftrue}; rpf \Leftarrow \text{if } ce_1 \text{ then } ce_2 \text{ else } ce_3 @ ce_4 \dashv \Delta' \rightsquigarrow (ce_1) \wedge \mathcal{C}} \text{RCHK_IFTRUE}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce_3 @ ce_4 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \text{iffalse}; rpf \Leftarrow \text{if } ce_1 \text{ then } ce_2 \text{ else } ce_3 @ ce_4 \dashv \Delta' \rightsquigarrow (\text{Not } ce_1) \wedge \mathcal{C}} \text{RCHK_IFFALSE}$$

$$\begin{array}{l}
\Sigma_c = \text{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \rightarrow_{\text{eff}} \tau' = ce_0 \in \Sigma_c \quad \text{eff} \leq \mathbf{S} \\
\Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \\
\Sigma_r; \Delta \vdash rpf \Leftarrow [\text{id}(\Gamma), (ce : \tau)/x] ce_0 @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C}
\end{array}$$

$$\frac{}{\Sigma_r; \Delta \vdash \text{unfold}; rpf \Leftarrow f ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RCHK_UNFOLD}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Leftarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash \text{annot}; rpf \Leftarrow (ce : \tau) @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RCHK_ANNOT}$$

$$\frac{\Sigma_r; \Delta \vdash rpf \Rightarrow ce_1 @ ce'_1 \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash rpf \Leftarrow ce_2 @ ce'_2 \dashv \Delta' \rightsquigarrow (\mathcal{C} \wedge (ce_1 \doteq ce_2)) \wedge (ce'_1 \doteq ce'_2)} \text{RCHK_SUB}$$

$$\boxed{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash \cdot = \cdot \rightsquigarrow \top} \text{PF EQ_NIL}$$

$$\frac{\Sigma_r; \Delta, x : \tau[\text{eff}] \vdash Pf_1 = [\text{id}(|\Delta|), x/y] Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash x :_{\text{eff}} \tau, Pf_1 = y :_{\text{eff}} \tau, Pf_2 \rightsquigarrow \forall x : \tau. \mathcal{C}} \text{PF EQ_COMP}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1 [\text{log}], Pf_1 = ce_2 [\text{log}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \mathcal{C}} \text{PF EQ_LOG}$$

$$\frac{\Sigma_r; \Delta \vdash Pf_1 = Pf_2 \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash ce_1 @ ce'_1 [\text{res}], Pf_1 = ce_2 @ ce'_2 [\text{res}], Pf_2 \rightsquigarrow (((ce_1 \doteq ce_2)) \wedge (ce'_1 \doteq ce'_2)) \wedge \mathcal{C}} \text{PF EQ_RES}$$

$$\begin{array}{l}
\Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \text{Pred } \tau \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_2 \Rightarrow \text{Pred } \tau \\
\Sigma_r; \Delta, z : \tau[\mathbf{S}] \vdash [\text{id}(|\Delta|), z/y] Pf_1 = [\text{id}(|\Delta|), z/w] Pf_2 \rightsquigarrow \mathcal{C}
\end{array}$$

$$\frac{}{\Sigma_r; \Delta \vdash (y).ce_1 [\text{res}], Pf_1 = (w).ce_2 [\text{res}], Pf_2 \rightsquigarrow ((ce_1 \doteq ce_2)) \wedge \forall z : \tau. \mathcal{C}} \text{PF EQ_DEPRES}$$

$$\boxed{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}$$

$$\frac{}{\Sigma_r \vdash \cdot \sqcap \cdot = \cdot} \text{MERGE_EMPTY}$$

$$\begin{array}{c}
\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x : \tau[eff] \sqcap \Delta_1, x : \tau[eff] = \Delta, x : \tau[eff]} \text{MERGE_EFF} \\
\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta}{\Sigma_r \vdash \Delta_0, x : ce[\log] \sqcap \Delta_1, x : ce[\log] = \Delta, x : ce[\log]} \text{MERGE_LOG} \\
\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta \quad u = u_0 \sqcap u_1}{\Sigma_r \vdash \Delta_0, x : ce@ce'[\text{res}(u_0)] \sqcap \Delta_1, x : ce@ce'[\text{res}(u_1)] = \Delta, x : ce@ce'[\text{res}(u)]} \text{MERGE_RES} \\
\boxed{\Sigma_r; \Delta \vdash_{eff} cpat : \tau \dashv \Delta' \rightsquigarrow ce} \\
\frac{eff' = \lfloor eff \rfloor}{\Sigma_r; \Delta \vdash_{eff} x : \tau \dashv \Delta, x : \tau[eff'] \rightsquigarrow x} \text{CPM_VAR} \\
\frac{eff' = \lfloor eff \rfloor}{\Sigma_r; \Delta \vdash_{eff} (x_1, \dots, x_n) : (\tau_1, \dots, \tau_n) \dashv \Delta, x_1 : \tau_1[eff'], \dots, x_n : \tau_n[eff'] \rightsquigarrow (x_1, \dots, x_n)} \text{CPM_TUPLE} \\
\boxed{\Sigma_r; \Delta \vdash lpat : ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \\
\frac{}{\Sigma_r; \Delta \vdash x : ce \dashv \Delta, x : ce[\log] \rightsquigarrow \top} \text{LPM_VAR} \\
\frac{}{\Sigma_r; \Delta \vdash \text{auto} : ce \dashv \Delta \rightsquigarrow ce} \text{LPM_AUTO} \\
\boxed{\Sigma_r; \Delta \vdash_{eff} rpat : ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \\
\frac{}{\Sigma_r; \Delta \vdash_{eff} x : ce@ce' \dashv \Delta, x : ce@ce'[\text{res}(1)] \rightsquigarrow \top} \text{RPM_VAR} \\
\frac{\Sigma_r; \Delta \vdash lpat : ce \doteq ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{eff} \text{return } lpat : \text{return } ce@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RPM_RETURN} \\
\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \text{Pred } \tau \\ \Sigma_r; \Delta_0 \vdash_{\mathbf{S}} cpat : \tau \dashv \Delta_1 \rightsquigarrow ce \\ \Sigma_r; \Delta_1 \vdash_{eff} rpat_1 : ce_1@ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{eff} rpat_2 : [\text{id}(\Gamma), ce/x]ce_2@ce' \dashv \Delta_3 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{eff} \text{take}(cpat, rpat_1); rpat_2 : \text{take } x = ce_1; ce_2@ce' \dashv \Delta_3 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \text{RPM_TAKE} \\
\frac{\Sigma_r; \Delta \vdash lpat : \text{false} \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{eff} \text{fail}[lpat] : \text{fail}@ce \dashv \Delta' \rightsquigarrow \mathcal{C}} \text{RPM_FAIL} \\
\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce_1 \Rightarrow \tau \\ \Sigma_r; \Delta_0 \vdash_{\mathbf{S}} cpat : \tau \dashv \Delta_1 \rightsquigarrow ce \\ \Sigma_r; \Delta_1 \vdash lpat : ce \doteq ce_1 \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{eff} rpat : [\text{id}(\Gamma), (ce : \tau)/x]ce_2@ce_3 \dashv \Delta_3 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{eff} \text{let}[lpat]cpat; rpat : \text{let } x = ce_1; ce_2@ce_3 \dashv \Delta_3 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \text{RPM_LET} \\
\frac{\Sigma_r; \Delta \vdash_{eff} rpat : ce_1@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{eff} \text{iftrue}; rpat : \text{if } ce \text{ then } ce_1 \text{ else } ce_2@ce' \dashv \Delta' \rightsquigarrow (ce) \wedge \mathcal{C}} \text{RPM_IFTRUE} \\
\frac{\Sigma_r; \Delta \vdash_{eff} rpat : ce_2@ce' \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{eff} \text{iffalse}; rpat : \text{if } ce \text{ then } ce_1 \text{ else } ce_2@ce' \dashv \Delta' \rightsquigarrow (\text{Not } ce) \wedge \mathcal{C}} \text{RPM_IFFALSE} \\
\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad eff' = \lfloor eff \rfloor \\ \Sigma_c; \Gamma \vdash_{eff'} ce \Rightarrow D(\tau_1, \dots, \tau_i) \\ L : \tau \in \text{ctors}(D) \in \Sigma_c \\ ce'' = \text{branch}(L, x, ce''') \\ \Sigma_r; \Delta_0 \vdash_{\mathbf{S}} cpat : \tau \dashv \Delta_1 \rightsquigarrow ce_0 \\ \Sigma_r; \Delta_1 \vdash lpat : (L ce_0) \doteq ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{eff} rpat : [\text{id}(\Gamma), (ce_0 : \tau)/x]ce''@ce' \dashv \Delta_3 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{eff} \text{case}[lpat]Lcpat; rpat : ce'''@ce' \dashv \Delta_3 \rightsquigarrow (\text{is } ce \ L) \wedge \mathcal{C}_1 \wedge \mathcal{C}_2} \text{RPM_CASE}
\end{array}$$

$$\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad f(\mathbf{x} : \tau) \rightarrow_{\text{eff}'} \tau' = ce_0 \in \Sigma_c \quad \text{eff}' \leq \mathbf{S} \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \\ \Sigma_r; \Delta \vdash_{\text{eff}} \text{rpat} : [\text{id}(\Gamma), (ce : \tau)/x] ce_0 @ ce' \vdash \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{unfold}; \text{rpat} : f \text{ } ce @ ce' \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_UNFOLD}$$

$$\frac{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rpat} : ce @ ce' \vdash \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{annot}; \text{rpat} : (ce : \tau) @ ce' \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RPM_ANNOT}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} q : Pf \vdash \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{}{\Sigma_r; \Delta \vdash_{\text{eff}} \cdot : \cdot \vdash \Delta \rightsquigarrow \top} \quad \text{QM_EMPTY}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta_0| \\ \Sigma_r; \Delta_0 \vdash_{\text{eff}'} \text{cpat} : \tau \vdash \Delta_1 \rightsquigarrow ce \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} q : [\text{id}(\Gamma), (ce : \tau)/x] Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} (\text{cpat}, q) : x :_{\text{eff}'} \tau, Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C}} \quad \text{QM_COMP_CPAT}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta_0 \vdash \text{lpat} : ce \vdash \Delta_1 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} q : Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} (\log \text{lpat}, q) : ce [\log], Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \text{QM_COMP_LOG}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{rpat} : ce @ ce' \vdash \Delta_1 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} q : Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} (\text{res } \text{rpat}, q) : ce @ ce' [\text{res}], Pf \vdash \Delta_2 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \text{QM_COMP_RES}$$

$$\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \text{Pred } \tau \\ \Sigma_r; \Delta_0 \vdash_{\mathbf{S}} \text{cpat} : \tau \vdash \Delta_1 \rightsquigarrow ce' \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} \text{rpat} : ce @ ce' \vdash \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\text{eff}} q : Pf \vdash \Delta_3 \rightsquigarrow \mathcal{C}_2 \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} (\text{res do } \text{cpat} = \text{rpat}, q) : (y).ce [\text{res}], Pf \vdash \Delta_3 \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_2} \quad \text{QM_COMP_DEPRES}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Rightarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_r; \Gamma \vdash_{\text{eff}} Pf \text{ wf} \\ \Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} : Pf \Rightarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSYN_ANNOT}$$

$$\frac{\begin{array}{l} \Sigma_r \vdash f : Pf_1 \multimap Pf_2 [\text{eff}'] \quad \text{eff}' \leq \text{eff} \quad \text{eff}'' = \lfloor \text{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\text{eff}''} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \vdash \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} f \text{ rsp} \Rightarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSYN_CALL}$$

$$\frac{\begin{array}{l} \text{prim} : Pf_1 \multimap Pf_2 [\text{eff}'] \quad \text{eff}' \leq \text{eff} \quad \text{eff}'' = \lfloor \text{eff} \rfloor \\ \Sigma_r; \Delta \vdash_{\text{eff}''} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \vdash \Delta' \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{prim } \text{rsp} \Rightarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSYN_PRIMAPP}$$

$$\frac{\Sigma_r; \Delta \vdash \text{rpf} \Rightarrow \text{take } x = ce_1; ce_2 @ ce_3 \vdash \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{open-take } \text{rpf} \Rightarrow x :_{\mathbf{S}} \tau, ce_1 @ x [\text{res}], ce_2 @ ce_3 [\text{res}], \cdot \vdash \Delta' \rightsquigarrow \mathcal{C}} \quad \text{RCSYN_OPENTAKE}$$

$$\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \text{type } D(a, b) = \{\text{Next}:a \mid \text{Done}:b\} \in \Sigma_c \\ \Gamma = |\Delta| \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Rightarrow \text{Pred } D(A, B) \\ \Sigma_r; \Delta \vdash_{\mathbf{0}} \text{crt}_1 \Leftarrow x :_{\mathbf{0}} A, ce @ \text{Next } x [\text{res}], \cdot \vdash \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta', x : A[\mathbf{0}], y : ce @ \text{Next } x [\text{res } (1)] \vdash_{\mathbf{1}} \text{crt}_2 \Leftarrow z :_{\mathbf{0}} D(A, B), ce @ z [\text{res}], \cdot \vdash \Delta', x : A[\mathbf{0}], y : ce @ \text{Next } x [\text{res } (0)] \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\mathbf{1}} \text{iter } [ce](q = \text{crt}_1) \{ \text{crt}_2 \} \Rightarrow z :_{\mathbf{0}} B, ce @ \text{Done } z [\text{res}], \cdot \vdash \Delta' \rightsquigarrow \mathcal{C} \wedge \forall x : A. \mathcal{C}'}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} y : ce \text{ is } Lx \rightarrow \text{crt} \Leftarrow Pf \vdash \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\begin{array}{l} \Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\text{eff}} ce \Rightarrow D(\tau_1, \dots, \tau_i) \\ L : \tau \text{ in } D(\tau_1, \dots, \tau_i) \text{ in } \Sigma_c \\ \Sigma_r; \Delta, x : \tau[\text{eff}], y : ce \doteq Lx[\log] \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \vdash \Delta', x : \tau[\text{eff}], y : ce \doteq Lx[\log] \rightsquigarrow \mathcal{C} \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} y : ce \text{ is } Lx \rightarrow \text{crt} \Leftarrow Pf \vdash \Delta' \rightsquigarrow \forall x : \tau. ce \doteq Lx \Rightarrow \mathcal{C}} \quad \text{CBR_BRANCH}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{crt}_1 \Rightarrow Pf' \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} q : Pf' \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\text{eff}} \text{crt}_2 \Leftarrow Pf \dashv \Delta_3, \Delta_4 \rightsquigarrow \mathcal{C}_2 \\ \text{len}(\Delta_1) = \text{len}(\Delta_3) \\ \text{zero}(\Delta_4) \\ \mathcal{C}' = \Delta_4 \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{let } q = \text{crt}_1; \text{crt}_2 \Leftarrow Pf \dashv \Delta_3 \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCCHK_LET}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta_0 \vdash \text{lpf} \Rightarrow ce \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta_1 \vdash \text{lpat} : ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta_3, \Delta_4 \rightsquigarrow \mathcal{C}_2 \\ \text{len}(\Delta_1) = \text{len}(\Delta_3) \\ \text{zero}(\Delta_4) \\ \mathcal{C}' = \Delta_4 \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{let log lpat} = \text{lpf}; \text{crt} \Leftarrow Pf \dashv \Delta_3 \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCCHK_LETLOG}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta_0 \vdash \text{rpf} \Rightarrow ce@ce' \dashv \Delta_1 \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta_1 \vdash_{\text{eff}} \text{rpat} : ce@ce' \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta_3, \Delta_4 \rightsquigarrow \mathcal{C}_2 \\ \text{len}(\Delta_1) = \text{len}(\Delta_3) \\ \text{zero}(\Delta_4) \\ \mathcal{C}' = \Delta_4 \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{let res rpat} = \text{rpf}; \text{crt} \Leftarrow Pf \dashv \Delta_3 \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCCHK_LETRES}$$

$$\frac{\begin{array}{l} \Sigma_c = \text{Comp}(\Sigma_r) \quad \Gamma = |\Delta_0| \quad \text{eff}' = \lfloor \text{eff} \rfloor \\ \Sigma_c; \Gamma \vdash_{\text{eff}'} ce \Rightarrow \tau \\ \Sigma_r; \Delta_0 \vdash_{\text{eff}'} \text{cpat} : \tau \dashv \Delta_1 \rightsquigarrow ce' \\ \Sigma_r; \Delta_1 \vdash \text{lpat} : ce' \doteq ce \dashv \Delta_2 \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta_2 \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta_3, \Delta_4 \rightsquigarrow \mathcal{C}_2 \\ \text{len}(\Delta_0) = \text{len}(\Delta_3) \\ \text{zero}(\Delta_4) \\ \mathcal{C} = \Delta_4 \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \end{array}}{\Sigma_r; \Delta_0 \vdash_{\text{eff}} \text{let core} [\text{lpat}] \text{cpat} = ce; \text{crt} \Leftarrow Pf \dashv \Delta_3 \rightsquigarrow \mathcal{C}} \quad \text{RCCHK_LETCORE}$$

$$\frac{\begin{array}{l} \text{eff}' = \lfloor \text{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\text{eff}'} ce \Leftarrow \text{Bool} \\ \Sigma_r; \Delta, x : ce \doteq \text{true}[\log] \vdash_{\text{eff}'} \text{crt}_1 \Leftarrow Pf \dashv \Delta_1, x : ce \doteq \text{true}[\log] \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta, x : ce \doteq \text{false}[\log] \vdash_{\text{eff}'} \text{crt}_2 \Leftarrow Pf \dashv \Delta_2, x : ce \doteq \text{false}[\log] \rightsquigarrow \mathcal{C}_2 \\ \Sigma_r \vdash \Delta_1 \sqcap \Delta_2 = \Delta' \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{if } [x] \text{ce then crt}_1 \text{ else crt}_2 \Leftarrow Pf \dashv \Delta' \rightsquigarrow (ce \doteq \text{true} \Rightarrow \mathcal{C}_1) \wedge (ce \doteq \text{false} \Rightarrow \mathcal{C}_2)} \quad \text{RCCHK_IF}$$

$$\frac{\begin{array}{l} \text{eff}' = \lfloor \text{eff} \rfloor \\ \Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\text{eff}'} ce \Rightarrow D(\tau_1, \dots, \tau_i) \\ D(a_1, \dots, a_k) = \{L_1 : \tau'_1 \mid \dots \mid L_n : \tau'_n\} \in \Sigma_c \\ \Sigma_r; \Delta \vdash_{\text{eff}'} y : ce \text{ is } L_1 x_1 \rightarrow \text{crt}_1 \Leftarrow Pf \dashv \Delta_1 \rightsquigarrow \mathcal{C}_1 \quad \dots \quad \Sigma_r; \Delta \vdash_{\text{eff}'} y : ce \text{ is } L_n x_n \rightarrow \text{crt}_n \Leftarrow Pf \dashv \Delta_n \rightsquigarrow \mathcal{C}_n \\ \Sigma_r \vdash \sqcap(\Delta_1; \dots; \Delta_n) = \Delta' \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{case } [y] \text{ce of } \{L_1 x_1 \rightarrow \text{crt}_1 \mid \dots \mid L_n x_n \rightarrow \text{crt}_n\} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}_1 \wedge \mathcal{C}_n} \quad \text{RCCHK}$$

$$\frac{\Delta' = \text{affinize}(\Delta)}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{exfalse} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \perp} \quad \text{RCCHK_EXFALSE}$$

$$\frac{\begin{array}{l} \Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Rightarrow Pf' \dashv \Delta' \rightsquigarrow \mathcal{C} \\ \Sigma_r; \Delta' \vdash Pf' = Pf \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RCCHK_SUB}$$

$$\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}$$

$$\begin{array}{c}
\overline{\Sigma_r; \Delta \vdash_{\text{eff}} \cdot : \cdot \multimap Pf \gg Pf \dashv \Delta \rightsquigarrow \top} \quad \text{SPINE_NIL} \\
\\
\frac{\Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\text{eff}} ce \Leftarrow \tau \quad \Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : [\text{id}(|\Delta|), ce/x]Pf_1 \multimap [\text{id}(|\Delta|), ce/x]Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\text{eff}} ce, \text{rsp} : x:\mathbf{0}\tau, Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{SPINE_PURE} \\
\\
\frac{\Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\mathbf{S}} ce \Leftarrow \tau \quad \Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : [\text{id}(|\Delta|), ce/x]Pf_1 \multimap [\text{id}(|\Delta|), ce/x]Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\text{eff}} ce, \text{rsp} : x:\mathbf{S}\tau, Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{SPINE_SPEC} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{lpf}, \text{rsp} : ce [\text{log}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{SPINE_LOG} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{rpf} \Leftarrow ce_1 @ ce_2 \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rpf}, \text{rsp} : ce_1 @ ce_2 [\text{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{SPINE_RES} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{rpf} \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : [\text{id}(|\Delta'|), ce''/y]Pf_1 \multimap [\text{id}(|\Delta'|), ce''/y]Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rpf}, \text{rsp} : (y).ce [\text{res}], Pf_1 \multimap Pf_2 \gg Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \text{SPINE_DEPRES} \\
\\
\boxed{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}
\end{array}$$

$$\begin{array}{c}
\overline{\Sigma_r; \Delta \vdash_{\text{eff}} \cdot : \cdot \dashv \Delta \rightsquigarrow \top} \quad \text{TUP_NIL} \\
\\
\frac{\Gamma = |\Delta| \quad \Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \Gamma \vdash_{\text{eff}} ce \Leftarrow \tau \quad \Sigma_r; \Delta \vdash_{\text{eff}} \text{rsp} : [\text{id}(|\Delta|), ce/x]Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r; \Delta \vdash_{\text{eff}} ce, \text{rsp} : x:\text{eff}\tau, Pf \dashv \Delta' \rightsquigarrow \mathcal{C}} \quad \text{TUP_CORE} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{lpf} \Leftarrow ce \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{lpf}, \text{rsp} : ce [\text{log}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{TUP_LOG} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{rpf} \Leftarrow ce @ ce' \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rpf}, \text{rsp} : ce @ ce' [\text{res}], Pf \dashv \Delta'' \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{TUP_RES} \\
\\
\frac{\Sigma_r; \Delta \vdash \text{rpf} \Rightarrow ce' @ ce'' \dashv \Delta' \rightsquigarrow \mathcal{C} \quad \Sigma_r; \Delta' \vdash_{\text{eff}} \text{rsp} : [\text{id}(|\Delta'|), ce''/y]Pf \dashv \Delta'' \rightsquigarrow \mathcal{C}'}{\Sigma_r; \Delta \vdash_{\text{eff}} \text{rpf}, \text{rsp} : (y).ce [\text{res}], Pf \dashv \Delta'' \rightsquigarrow (\mathcal{C} \wedge (ce \doteq ce')) \wedge \mathcal{C}'} \quad \text{TUP_DEPRES} \\
\\
\boxed{\Sigma_r \vdash \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}
\end{array}$$

$$\begin{array}{c}
\frac{\Sigma_r; \cdot \vdash_{\text{eff}} \text{crt} \Leftarrow Pf \dashv \Delta' \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{main} : Pf_{\text{eff}} = \text{crt} \dashv \Sigma_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_MAIN} \\
\\
\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c \vdash \text{sort } D(\bar{a}) = \{\bar{L} : \tau\} \text{ ok} \quad \Sigma_r, \text{sort } D(\bar{a}) = \{\bar{L} : \tau\} \vdash \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{sort } D(a_1, \dots, a_i) = \{L_1 : \tau_1 \dots L_n : \tau_n\} \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_SORTDECL} \\
\\
\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c \vdash \text{type } D(\bar{a}) = \{\bar{L} : A\} \text{ ok} \quad \Sigma_r, \text{type } D(\bar{a}) = \{\bar{L} : A\} \vdash \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{type } D(a_1, \dots, a_i) = \{L_1 : A_1 \dots L_n : A_n\} \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_TYPEDECL} \\
\\
\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c; \cdot, x:\mathbf{0}A \vdash_{\mathbf{0}} ce \Leftarrow B \quad \Sigma_r, \text{fun } f(x : A) \rightarrow_{\mathbf{0}} B = ce \vdash \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{fun } f(x : A) \rightarrow_{\mathbf{0}} B = ce \quad \text{rprog} \dashv \Sigma'_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_PUREFUN}
\end{array}$$

$$\begin{array}{c}
\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c, f : A \rightarrow_1 B; \cdot, x :_0 A \vdash_1 ce \Leftarrow B \quad \Sigma_r, f : A \rightarrow_1 B \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{fun } f(x : A) \rightarrow_1 B = ce \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_IMPUREFUN} \\
\\
\frac{\Sigma_c = \text{Comp}(\Sigma_r) \quad \Sigma_c, f : \tau \rightarrow_{\mathbf{S}} \tau'; \cdot, x :_{\mathbf{S}} \tau \vdash_{\mathbf{S}} ce \Leftarrow \tau' \quad \Sigma_r, \text{fun } f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}}{\Sigma_r \vdash \text{fun } f(x : \tau) \rightarrow_{\mathbf{S}} \tau' = ce \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}} \quad \text{RPROG_SPECFUN} \\
\\
\frac{\begin{array}{l} \Sigma_r \vdash Pf_1 \multimap Pf_2 [0] \text{ wf} \\ \Sigma_r; \cdot \vdash_0 q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r; \Delta \vdash_0 crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \text{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2 [0] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r \vdash \text{fun } f(q) : Pf_1 \multimap Pf_2 [0] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RPROG_PUREREFINED} \\
\\
\frac{\begin{array}{l} \Sigma_r \vdash Pf_1 \multimap Pf_2 [1] \text{ wf} \\ \Sigma_r; \cdot \vdash_0 q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r, f : Pf_1 \multimap Pf_2 [1]; \Delta \vdash_1 crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \text{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2 [1] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r \vdash \text{fun } f(q) : Pf_1 \multimap Pf_2 [1] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RPROG_IMPUREREFINED} \\
\\
\frac{\begin{array}{l} \Sigma_r \vdash Pf_1 \multimap Pf_2 [\mathbf{S}] \text{ wf} \\ \Sigma_r; \cdot \vdash_{\mathbf{S}} q : Pf_1 \dashv \Delta \rightsquigarrow \mathcal{C}_1 \\ \Sigma_r, f : Pf_1 \multimap Pf_2 [\mathbf{S}]; \Delta \vdash_{\mathbf{S}} crt \Leftarrow Pf_2 \dashv \Delta' \rightsquigarrow \mathcal{C}_2 \\ \text{zero}(\Delta') \\ \mathcal{C} = \Delta' \Rightarrow (\mathcal{C}_1 \wedge \mathcal{C}_2) \\ \Sigma_r, f : Pf_1 \multimap Pf_2 [\mathbf{S}] \vdash rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C}' \end{array}}{\Sigma_r \vdash \text{fun } f(q) : Pf_1 \multimap Pf_2 [\mathbf{S}] = crt \quad rprog \dashv \Sigma'_r \rightsquigarrow \mathcal{C} \wedge \mathcal{C}'} \quad \text{RPROG_SPECREFINED}
\end{array}$$

$$\boxed{\Sigma_r \vdash \sqcap \overline{\Delta} = \Delta}$$

$$\overline{\Sigma_r \vdash \sqcap \Delta_0; \cdot = \Delta_0} \quad \text{NMERGE_BASE}$$

$$\frac{\Sigma_r \vdash \Delta_0 \sqcap \Delta_1 = \Delta' \quad \Sigma_r \vdash \sqcap(\Delta'; \overline{\Delta}) = \Delta}{\Sigma_r \vdash \sqcap(\Delta_0; \Delta_1; \overline{\Delta}) = \Delta} \quad \text{NMERGE_STEP}$$

Definition rules: 302 good 0 bad
Definition rule clauses: 707 good 0 bad